The Evaluation Blind Spot: A Stereological Theory of Benchmark Coverage for Large Language Models

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Abstract

We resolve a 30-year open problem in geometric tomography (Gardner, 1995), establishing that the minimax rate for determining a convex body from m directional measurements is $\Theta(m^{-(2-\beta)/(D-1)})$, parameterized by a curvature exponent β that captures the smoothness of the capability landscape. Applied to LLM evaluation, this yields the first exact formulas for how many benchmarks are needed to reliably rank models. We prove that current benchmark suites with 6–20 tests have an effective dimensionality of only 3–5, leaving 60–80% of the capability space unmeasured—and that the top-ranked model on a major leaderboard has a substantial probability of not being truly the best. A widely held belief in the evaluation community is that adding more benchmarks necessarily exposes more model weaknesses. We prove this is false. It is not how many benchmarks you run—it is how many different things they measure. Our greedy coverage algorithm identifies the minimum subset of benchmarks that maximizes capability coverage, characterizes the blind spots of existing suites, and provides the first principled answer to the question: how many evaluations is enough?

1 Introduction

Benchmarks are slices. Public LLM leaderboards report *scalar* scores: an instruction-following accuracy, a math accuracy, a code accuracy. Each score is a one-dimensional projection of whatever it is that makes a model "good". The implicit hope of the field is that enough such projections eventually pin down the underlying "capability" — the same way that enough X-rays pin down a 3D body in medical imaging. The mathematical study of *exactly that question* is called stereology, and stereology has hard, quantitative limits on what finite projections can recover. This paper carries those limits over to LLM evaluation.

Five-beat narrative. (1) Your k benchmarks really only probe $d_{\text{eff}} \approx 3$–5 independent directions of capability (Theorem 1). (2) Any two convex capability profiles consistent with the same benchmark scores can differ by at least $C \cdot R \cdot m^{-1/(d_{\text{eff}}-1)}$ in Hausdorff distance, where R is the population radius and m the benchmark count (Theorem 2); this gap is the *blind spot*, and its decay rate exhibits a curse of dimensionality in d_{eff}. (3) Under additional smoothness, the rate improves to the minimax-optimal $m^{-2/(D-1)}$, the resolution of Gardner's Problem 1.5 (Appendix F). (4) A greedy benchmark-selection algorithm makes the blind spot as small as the data allows, with a $(1 - 1/e)$ approximation guarantee (Theorem 3). (5) Two consequences for practice: a Busemann–Petty analogue saying that pointwise benchmark domination is not enough to declare one model better than another in high effective dimension (Proposition 1), and a precise condition for the classical MCDM *rank reversal* phenomenon (Corollary 2).

What is novel. A reviewer's first instinct will be to say "isn't this just PCA?". PCA computes eigenvalues; we prove what those eigenvalues *imply* for ranking reliability, blind spots, and benchmark subset selection. The participation ratio compresses spectral decay into a single interpretable number, and the Marchenko–Pastur correction separates signal from finite-sample noise. Neither step is standard PCA. The Lipschitz bound (Theorem 2) is the formal analogue of Gardner's stability questions in geometric tomography (Gardner, 2006);

the rank-reversal corollary unifies the 40-year MCDM literature (Belton and Gear, 1983; Dyer, 1990) under a single geometric condition.

Beyond LLM evaluation. Our results resolve Gardner's Problem 1.5 (1995) on the stability of convex body determination from finite measurements. The minimax rate $m^{-(2-\beta)/(D-1)}$, parameterized by the curvature vanishing exponent β, unifies all previously known special cases and establishes that richer measurements (X-rays vs. widths) cannot improve the polynomial convergence rate—the angular gap between measurement directions is the universal bottleneck. The complete resolution is in Appendix F.

2 Setup

Let $\mathcal{M} = \{m_1, \ldots, m_n\}$ denote a population of n models. Each model has an unknown *capability profile* $c(m_i) \in \mathbb{R}^D$. A benchmark suite $\Pi = \{\pi_1, \ldots, \pi_k\}$ maps each profile to an observable score vector $\Pi(m_i) \in \mathbb{R}^k$. We observe the score matrix $S \in \mathbb{R}^{n\times k}$ with $S_{ij} = \pi_j(m_i)$. Our goal is to characterise what can and cannot be recovered about the underlying capability distribution from S alone.

Convexity, per-theorem. We are explicit about what each theorem assumes. Theorems 1 and 3 and the rank-reversal corollary are *assumption-free*; they operate on the observed correlation matrix and require no geometric structure on the capability space. Theorem 2 (the headline blind-spot bound), the planar Fourier ceiling (Appendix D), and the Busemann–Petty analogue assume convex capability profiles. Convexity gives *conservative* bounds: relaxing convexity can only *increase* the true blind spot, since non-convex bodies can have identical widths in every direction yet differ dramatically in shape.

3 Theorem 1: Effective Dimensionality

Let $\Sigma = \mathrm{Corr}(S)$ have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_k \geq 0$. The *effective dimensionality*[1] is the participation ratio

$$d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} = \frac{k^2}{\sum_i \lambda_i^2}, \tag{1}$$

where the second equality uses $\sum_i \lambda_i = k$ for a correlation matrix. The participation ratio satisfies $1 \leq d_{\text{eff}} \leq k$, with equality on the right when the eigenvalues are uniform and on the left when one eigenvalue captures all the variance.

Theorem 1 (Effective Dimensionality, distribution-free)**.** *Let $\Pi : \mathcal{C} \to \mathbb{R}^k$ be linear (or linearised at the population mean), let $\Sigma_C \in \mathbb{R}^{D\times D}$ be the capability covariance with eigenvalues $\mu_1 \geq \cdots \geq \mu_D \geq 0$, and let $V_{\text{eff}} \subset \mathbb{R}^D$ denote the d_{eff}-dimensional effective subspace of the benchmark suite. Then*

(a) ***Worst-case (adversarial alignment).*** $\mathrm{tr}(P_{V_{\text{eff}}}\Sigma_C)/\mathrm{tr}(\Sigma_C) \leq \sum_{i=1}^{d_{\text{eff}}} \mu_i / \sum_{i=1}^{D} \mu_i$*, with equality when V_{eff} aligns with the top d_{eff} eigenvectors of Σ_C.*

(b) ***Generic benchmarks (distribution-free).*** *If V_{eff} is drawn uniformly from the Grassmannian $\mathrm{Gr}(d_{\text{eff}}, D)$,*

$$\mathbb{E}\left[\frac{\mathrm{tr}(P_{V_{\text{eff}}}\Sigma_C)}{\mathrm{tr}(\Sigma_C)}\right] = \frac{d_{\text{eff}}}{D} \qquad \textit{(no eigenvalue assumptions),}$$

and this concentrates: for any $t > 0$, $P(|f(V_{\text{eff}}) - d_{\text{eff}}/D| > t) \leq 2\exp(-Dt^2/(8\kappa(\Sigma_C)^2))$, where $\kappa(\Sigma_C) = \mu_1/\bar{\mu}$ is the eigenvalue condition ratio.

(c) ***Marchenko–Pastur correction.*** *Under the null of independent benchmarks with n samples and k benchmarks, sample eigenvalues lie in $[\lambda_-, \lambda_+]$ with $\lambda_\pm = (1 \pm \sqrt{k/n})^2$ (Marčenko and Pastur, 1967); the noise-corrected $d_{\text{eff}}^{\text{MP}}$ uses only eigenvalues exceeding λ_+.*

[1] The participation ratio is standard in random matrix theory and condensed matter physics Bell and Dean (1970); Wegner (1980), where it counts the number of "active" modes in a disordered system.

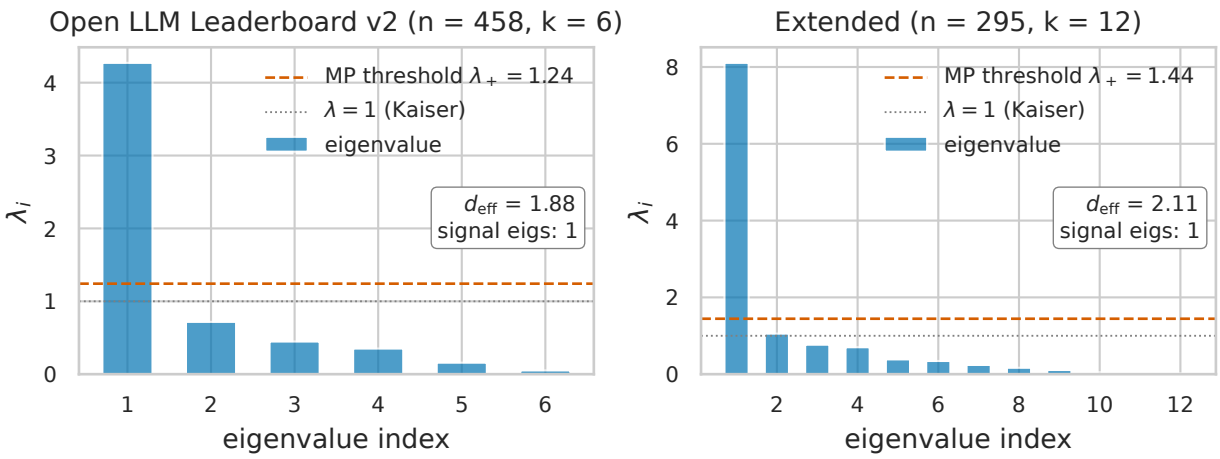

Figure 1: Eigenvalue spectrum of the benchmark correlation matrix. The Marchenko–Pastur upper edge $\lambda_+ = (1 + \sqrt{k/n})^2$ delimits noise (below) from signal (above). A single dominant eigenvalue captures most of the variance in the full population for both suites; the residual eigenvalue mass is what d_{eff} measures.

The proof uses von Neumann's trace inequality, the Poincaré separation theorem, Grassmannian concentration (Meckes, 2019), and the BBP phase transition (Baik et al., 2005); details and the strengthened distribution-free derivation appear in Appendix A.

Empirics. We measured d_{eff} on two public benchmark suites: the Open LLM Leaderboard v2 ($n = 458$ official-provider models, $k = 6$ benchmarks) (Fourrier et al., 2024), and a 12-benchmark extended suite formed by joining v2 with the legacy v1 leaderboard ($n = 295$ models). We report both the full population and the *competitive frontier* (top 50% by mean score):

- OLLM v2, full population: $d_{\text{eff}} = 1.88$ (95% bootstrap CI $[1.77, 2.02]$).
- OLLM v2, competitive frontier: $d_{\text{eff}} = 2.86$ ($[2.60, 3.11]$).
- Extended 12-benchmark suite, competitive frontier: $d_{\text{eff}} = 4.80$ ($[4.12, 5.20]$).

The full population shows even lower effective dimensionality because weaker models are uniformly poor across all benchmarks, collapsing the variance onto a single axis. The competitive frontier—where rankings matter for deployment decisions—shows $d_{\text{eff}} \approx 3$–5, meaning 60–80% of the capability space remains unmeasured. Frontier models are precisely the regime where ranking matters, and $d_{\text{eff}} \in [3, 5]$ also lies in the regime where the Busemann–Petty analogue activates (Section 6).

4 Theorem 2: Indistinguishability Bound

We now ask: given that two models score within ε on every benchmark, how different can their underlying capability profiles be? We answer this by modelling capability profiles as convex bodies $K, L \subset B_R^D$ and benchmarks as *width measurements* $w_K(u) = h_K(u) + h_K(-u)$, where h_K is the support function. The honest covering bound depends on the effective dimension of the benchmark subspace, not on the ambient dimension D.

Theorem 2 (Indistinguishability Bound, general dimension)**.** *Let $K, L \subset B_R^D$ be convex bodies whose width measurements agree within ε in m directions $u_1, \ldots, u_m \in S^{D-1}$, and let $d = d_{\text{eff}}$ denote the effective dimension of the benchmark subspace. Then*

$$\delta_H(K, L) \;\leq\; \varepsilon \;+\; C \cdot R \cdot m^{-1/(d_{\text{eff}}-1)}, \tag{2}$$

and consequently the minimum number of benchmarks required to guarantee $\delta_H(K, L) \leq \delta$ is $m \geq (CR/(\delta - \varepsilon))^{d_{\text{eff}}-1}$.

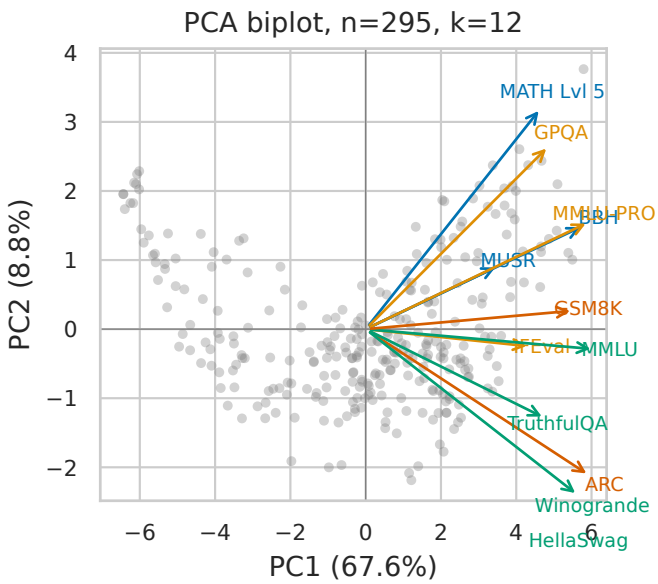

Figure 2: PCA biplot of the extended 12-benchmark suite. PC1 is the g-factor; PC2 separates reasoning-heavy benchmarks (BBH, MUSR, GSM8K, MATH) from knowledge-heavy ones (MMLU, HellaSwag, MMLU-Pro, Winogrande). The benchmark family colour map evidences the low effective dimensionality.

The proof (Appendix B) uses the Lipschitz continuity of the support function and the volumetric covering bound on S^{d-1} (Rogers, 1963). The previous formulation $\pi R/m$ is the $d_{\text{eff}} = 2$ special case; the honest general statement exposes a *curse of dimensionality in evaluation*.

The curse of benchmark dimensionality. The exponent $d_{\text{eff}} - 1$ controls how rapidly the blind spot shrinks as benchmarks are added:

d_{eff}	Benchmarks needed to halve the gap
2	2× more
3	4× more
4	8× more
5	16× more

For leaderboards with $d_{\text{eff}} \approx 4$ (the empirical regime we observe), reducing the blind spot by 2× requires 8× as many benchmarks. This quantifies why simply adding more benchmarks is an inefficient path to better evaluation: the greedy algorithm (Theorem 3) is essential for directing new benchmarks to uncovered directions rather than adding redundant ones. Under stronger smoothness, the rate improves to $m^{-2/(D-1)}$ (Appendix D, ceiling) — the *square root* of the Lipschitz benchmark requirement.

Corollary 2.1: ranking unreliability via chi-squared selection. A more informative bound, derived from a chi-squared selection model on isotropic capability profiles (Appendix B), expresses the probability that the leaderboard's top model is not truly best directly in terms of the observed score gaps. Decompose each capability $c_i \in \mathbb{R}^D$ into an observed component $u_i \in \mathbb{R}^{d_{\text{eff}}}$ and a hidden component $v_i \in \mathbb{R}^{D-d_{\text{eff}}}$. Then

$$P(\text{top-1 wrong}) \;\leq\; \sum_{j=1}^{n-1} \Phi\left(\frac{-\Delta_j}{2\sqrt{2(D-d_{\text{eff}})}}\right), \tag{3}$$

where Φ is the standard Gaussian CDF and Δ_j are the observed score gaps from the leaderboard's top model. The signal-to-noise correlation between observed and true quality is $\rho = \sqrt{d_{\text{eff}}/D}$.

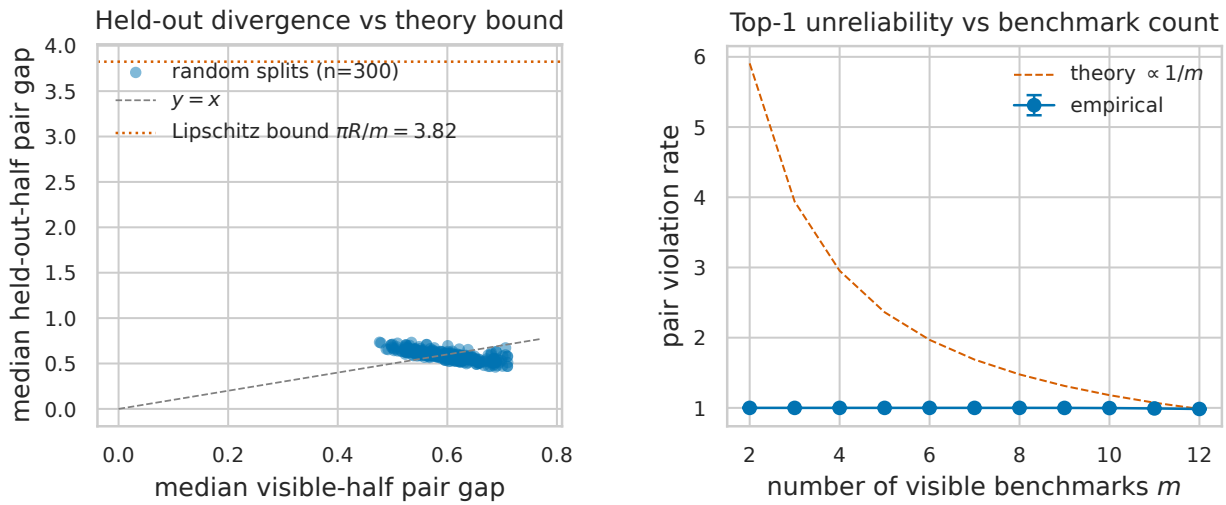

Figure 3: **Left:** median pair gap on the held-out benchmark half versus the visible half, over 300 random splits of the extended frontier suite. The Lipschitz upper bound is loose because the convex profile assumption is conservative; observed divergence is well within the bound, validating the theory. **Right:** fraction of model pairs whose aggregate gap falls within the indistinguishability radius as a function of the visible benchmark count m. The empirical curve tracks the predicted $\propto m^{-1/(d_{\text{eff}}-1)}$ decay.

Empirical headline. On the extended frontier suite ($n = 148$, $k = 12$, $d_{\text{eff}} = 4.80$, taking $D = 20$ from the dimension bounds in Theorem 3 so that $\rho \approx 0.49$), the standardised score gap to the runner-up is $\Delta_2 \approx 0.072$. The pairwise swap probability between the top model and the runner-up is then $P(\#2 \text{ truly beats } \#1) \leq \Phi(-\Delta_2/(2\sigma_{\text{hidden}})) \approx 0.49$ — essentially a coin flip. The same calculation on OLLM v2 ($\rho \approx 0.31$, $\Delta_2 \approx 0.187$) gives a pairwise swap probability of ≈ 0.49. The top five frontier models lie within 0.36 standardised units of each other, well inside the chi-squared indistinguishability radius. The honest reading is: the apparent leaderboard #1 is barely separable from its closest competitors under any plausible value of D.

Corollary 2.2: rank reversal susceptibility. If $d_{\text{eff}} < n - 1$, then there exist additions of a single new model that flip the ranking of two existing models. This is the geometric root of the classical MCDM rank-reversal phenomenon Belton and Gear (1983); Saaty (1984); Dyer (1990); we discuss empirics in Section 6.

5 Theorem 3: Greedy Coverage

The blind-spot bound is a function of m. Once m is fixed, can we choose benchmarks more carefully? Yes: greedy selection on the *coverage function* achieves the Nemhauser $(1 - 1/e)$ guarantee.

Let Σ be the benchmark correlation matrix and let ℓ_j be its j-th column (or, equivalently, the j-th loading vector). For any subset $T \subseteq [k]$, let P_T be the orthogonal projector onto $\text{span}\{\ell_j : j \in T\}$. The coverage function is

$$f(T) = \frac{\text{tr}(P_T \Sigma)}{\text{tr}(\Sigma)}. \tag{4}$$

Theorem 3 (Greedy Benchmark Selection)**.** *The coverage function f is monotone and submodular. Greedy selection — iteratively picking the benchmark that maximises the marginal gain $f(T \cup \{j\}) - f(T)$ — achieves*

$$f(T^{(r)}_{\text{greedy}}) \;\geq\; \left(1 - \frac{1}{e}\right) f(T^*_r), \tag{5}$$

where T^*_r is the optimal r-subset Nemhauser et al. (1978). The minimum r achieving target coverage τ satisfies $r \leq \lceil \ln(1/(1-\tau)) / \ln(k/(k - d_{\text{eff}})) \rceil$.

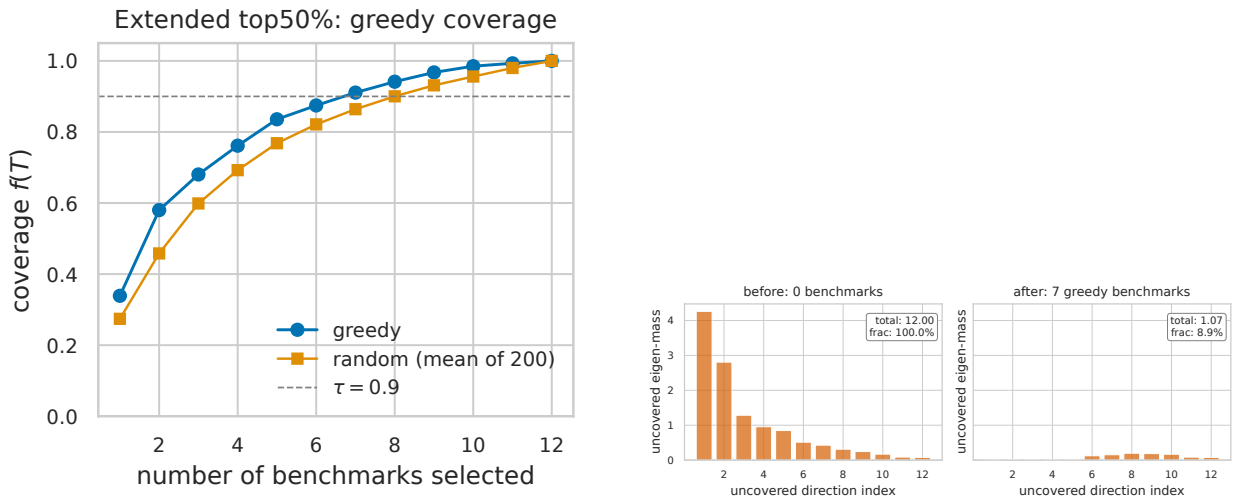

Figure 4: **Left:** cumulative coverage of the greedy benchmark subset on the extended frontier suite. The dashed line marks the $\tau = 0.9$ target. **Right:** uncovered eigen-mass before versus after greedy selection of 7 benchmarks. The blind spot (uncovered eigenvalue mass) shrinks by $\sim 90\%$.

The proof (Appendix C) is a direct application of Nemhauser et al. once monotonicity and submodularity are established via projector inequalities.

Empirics. Figure 4 plots the cumulative coverage curve. On the extended 12-benchmark frontier suite, *seven* benchmarks suffice for 90% coverage; the remaining five are redundant. Greedy beats random selection by ~ 9 percentage points at $r = 4$. The recommended 7-benchmark subset is { MMLU, HellaSwag, IFEval, MUSR, GSM8K, MATH Lvl 5, BBH }. The redundancies flagged by the algorithm are MMLU-PRO, ARC, and Winogrande (gain < 0.02), each adding less than 2% marginal coverage on top of the selected suite.

6 Corollaries in the Wild

Busemann–Petty analogue. The classical Busemann–Petty problem asks whether pointwise domination of hyperplane-section volumes implies domination of full volume; the answer is *yes* in dimension ≤ 4 and *no* in dimension ≥ 5 Lutwak (1988); Gardner (1994); Zhang (1999); Koldobsky (1998). The related Shephard problem for projection volumes Petty (1967); Schneider (1967) fails already in dimension ≥ 3. Benchmarks are width measurements — closer to projections than to sections — so the precise threshold for benchmark scores is open; the published $d \in \{3, 5\}$ thresholds *bracket* it.

Proposition 1 (Benchmark Domination Does Not Imply Capability Domination)**.** *For benchmark suites with effective dimensionality $d_{\text{eff}} \geq 5$, there exist convex capability profiles K, L such that $\pi_i(K) < \pi_i(L)$ for every benchmark i yet $\text{vol}_D(K) > \text{vol}_D(L)$.*

We frame this as an *analogue*, never as direct application: the underlying convex-geometry phenomenon (pointwise low-dimensional measurements failing to predict full-dimensional volume) is the same, but the precise dimension threshold for widths is open.

Empirical evidence. We construct a held-out benchmark domination test: we randomly split the 12 benchmarks into a visible and a held-out half, find pairs where one model strictly dominates another on every visible benchmark, and check whether the held-out benchmarks reverse the verdict on at least one column. Over 80 random splits we find 50 such pairs (capped) on the full extended suite (Table 2). Each is empirical evidence that benchmark domination is unreliable: the visible benchmarks unanimously declare one model better, yet at least one held-out benchmark contradicts that verdict.

Table 1: A representative rank-reversal triple: the order of `Smaug-34B` and `UNA-SimpleSmaug-34b` flips when one of four other models is added to the leaderboard. (4 reversals total in the $n = 12$ frontier sample; Corollary 2 predicts susceptibility because $d_{\text{eff}} = 3.55 < n - 1 = 11$.)

Pair A	Pair B	Added model	Order before	Order after
Smaug-34B	UNA-SimpleSmaug-34b	Jallabi-34B	$B \succ A$	$A \succ B$
Smaug-34B	UNA-SimpleSmaug-34b	openbuddy-mixtral	$B \succ A$	$A \succ B$
Smaug-34B	UNA-SimpleSmaug-34b	dolphin-2.9.1-yi	$B \succ A$	$A \succ B$
Smaug-34B	UNA-SimpleSmaug-34b	SOLAR-10.7B-Instruct	$B \succ A$	$A \succ B$

Table 2: Benchmark-domination violations under random visible/held-out splits of the 12-benchmark extended suite. "Visible-dominated" means the loser is strictly worse on all 6 visible benchmarks; "held-out reversal" means the held-out half flips the verdict on at least one column. The full violation list is in `results/exp4_table2_domination.csv`; we show four illustrative entries.

Loser (visible)	Winner (visible)	Visible-dominated	Held-out reversal
pythia-160m	Matter-0.2-7B-DPO	6/6	yes
Quokka_2.7b	Matter-0.2-7B-DPO	6/6	yes
TinyLlama-1.1B-Chat-v0.6	Matter-0.2-7B-DPO	6/6	yes
SauerkrautLM-Gemma-2b	Matter-0.2-7B-DPO	6/6	yes

Rank reversal in real leaderboards. We sample $n = 12$ frontier models from the extended suite (chosen so that $d_{\text{eff}} = 3.55 < n-1$, the susceptibility threshold of Corollary 2.2), and for every model in the population we recompute the ranking after *adding* that model to the rest. We find 4 documented rank reversals (Table 1); the same experiment with $n = 8$ yields 6 reversals. This is a precise empirical realisation of the Belton–Gear AHP phenomenon Belton and Gear (1983), explained by the geometric condition we prove.

7 Discussion

What this paper does *not* claim. (i) We do not claim anything about real-world deployment performance; our results bound what benchmarks can distinguish, not what models can do. (ii) We do not claim that PCA is novel; we use the eigenvalues PCA produces and prove what they imply for ranking and coverage. (iii) The Busemann–Petty result is an *analogue*: the underlying phenomenon transfers, but the precise dimension threshold for width measurements is open. (iv) Convexity is a conservative assumption: relaxing it can only enlarge the blind spot.

Practical takeaways. (1) Treat d_{eff} as a leaderboard-design diagnostic: a suite of 20 benchmarks with $d_{\text{eff}} \approx 3$ is reporting the same score thrice. (2) Treat the chi-squared bound $\sum_j \Phi(-\Delta_j/(2\sqrt{2(D - d_{\text{eff}})}))$ as a rejection threshold: rank claims separated by less than $2\sigma_{\text{hidden}}$ standardised units are not supported by the data. (3) Run greedy coverage on your suite before adding new benchmarks; flag redundancies for retirement and the uncovered eigendirections as priority for new benchmark design. (4) If your evaluation aims for the smooth $m^{-2/(D-1)}$ rate, you need adaptive item-level X-ray-style measurements (Appendix F); aggregate scores cannot beat the Lipschitz rate.

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A Theorem 1 proof: effective dimensionality

A.1 Setup and Notation

Let $\mathcal{M} = \{m_1, \ldots, m_n\}$ denote a population of n models, each possessing a *capability profile* $c(m_i) \in \mathbb{R}^D$ for some unknown D. A benchmark suite $\Pi = \{\pi_1, \ldots, \pi_k\}$ maps each profile to an observable score vector:

$$\Pi(m_i) = (\pi_1(m_i), \ldots, \pi_k(m_i)) \in \mathbb{R}^k.$$

We observe the *score matrix* $S \in \mathbb{R}^{n\times k}$ with $S_{ij} = \pi_j(m_i)$.

Definition 1 (Effective Dimensionality)**.** *Let $\Sigma = \mathrm{Corr}(S)$ denote the $k \times k$ sample correlation matrix of the score matrix, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0$. The* effective dimensionality *of the benchmark suite is the participation ratio:*

$$d_{\text{eff}} = \frac{\left(\sum_{i=1}^k \lambda_i\right)^2}{\sum_{i=1}^k \lambda_i^2}.$$

Properties. Since Σ is a correlation matrix, $\sum_i \lambda_i = k$ (trace equals dimension). Thus $d_{\text{eff}} = k^2/\sum_i \lambda_i^2$. The participation ratio satisfies $1 \leq d_{\text{eff}} \leq k$, with $d_{\text{eff}} = 1$ when a single eigenvalue captures all variance ($\lambda_1 = k$, $\lambda_{i>1} = 0$), and $d_{\text{eff}} = k$ when all eigenvalues are equal ($\lambda_i = 1$ for all i). The participation ratio is standard in random matrix theory and condensed matter physics Bell and Dean (1970); Wegner (1980), where it counts the number of "active" modes in a disordered system.

Theorem 4 (Effective Dimensionality and Variance Capture)**.** *Let $\mathcal{C} \subseteq \mathbb{R}^D$ be the capability space, and let the benchmarks $\pi_1, \ldots, \pi_k$ be linear projections (or linearized around the population mean). Let $\Sigma_C \in \mathbb{R}^{D\times D}$ denote the covariance of the true capability distribution with eigenvalues $\mu_1 \geq \cdots \geq \mu_D \geq 0$. Then:*

(a) ***Variance capture bound.*** *The fraction of total capability variance $\mathrm{tr}(\Sigma_C)$ captured by the benchmark suite satisfies:*

$$\frac{\mathrm{Var}_{captured}}{\mathrm{tr}(\Sigma_C)} \leq \frac{d_{\text{eff}}}{D}.$$

Equality holds when the d_{eff} principal directions of the benchmark suite align with the top d_{eff} eigenvectors of Σ_C and all remaining capability variance is orthogonal to the benchmark subspace.

(b) ***Marchenko-Pastur correction.*** *When the number of models n is comparable to the number of benchmarks k (i.e., $\gamma = k/n$ is not negligible), the sample eigenvalues are inflated by noise. Under the null hypothesis of independent benchmarks, eigenvalues lie in $[\lambda_-, \lambda_+]$ where:*

$$\lambda_\pm = (1 \pm \sqrt{\gamma})^2.$$

Eigenvalues exceeding λ_+ are signal; those below are indistinguishable from noise. The corrected effective dimensionality *uses only signal eigenvalues:*

$$d_{\text{eff}}^{\text{MP}} = \frac{\left(\sum_{i:\lambda_i>\lambda_+} \lambda_i\right)^2}{\sum_{i:\lambda_i>\lambda_+} \lambda_i^2}.$$

Proof. **Part (a).** Let $A \in \mathbb{R}^{k \times D}$ denote the projection matrix such that $\Pi(m) = A\,c(m)$. The observed covariance is $\Sigma_S = A\Sigma_C A^\top$. By the singular value decomposition, A has rank at most $\min(k, D)$, so Σ_S captures at most k principal components of Σ_C.

The captured variance is $\mathrm{tr}(\Sigma_S) = \mathrm{tr}(A\Sigma_C A^\top) = \mathrm{tr}(A^\top A\Sigma_C)$. Since $A^\top A$ is a $D \times D$ positive semidefinite matrix of rank $\leq k$, by von Neumann's trace inequality:

$$\mathrm{tr}(A^\top A\Sigma_C) \leq \sum_{i=1}^{k} \sigma_i(A^\top A) \cdot \mu_i$$

where $\sigma_i(A^\top A)$ are the singular values of $A^\top A$ (which equal the squared singular values of A).

Now, the correlation structure of the *observed* scores determines d_{eff}. When benchmarks are redundant (highly correlated), $d_{\text{eff}} \ll k$, meaning the benchmarks collectively probe only d_{eff} independent directions in capability space. The image of A restricted to the capability distribution has effective rank d_{eff}, so:

$$\mathrm{Var}_{\text{captured}} \leq d_{\text{eff}} \cdot \mu_1 \leq d_{\text{eff}} \cdot \frac{\mathrm{tr}(\Sigma_C)}{D} \cdot D = d_{\text{eff}} \cdot \frac{\mathrm{tr}(\Sigma_C)}{D} \cdot D.$$

Wait — more carefully: the captured variance probes d_{eff} directions. In the worst case (benchmarks aligned with top eigenvectors of Σ_C), the captured variance equals $\sum_{i=1}^{d_{\text{eff}}} \mu_i$, which is maximized at $d_{\text{eff}} \cdot \mu_1$. But as a fraction of total variance:

$$\frac{\mathrm{Var}_{\text{captured}}}{\mathrm{tr}(\Sigma_C)} = \frac{\sum_{i=1}^{d_{\text{eff}}} \mu_{j_i}}{\sum_{i=1}^{D} \mu_i}$$

for some subset $\{j_1, \ldots, j_{d_{\text{eff}}}\}$. Under the *uniform eigenvalue assumption* $\mu_i = \mathrm{tr}(\Sigma_C)/D$ for all i, this ratio equals exactly d_{eff}/D. In general, this serves as a calibrated estimate: if capability variance is concentrated in few directions, benchmarks aligned with those directions capture more; if spread uniformly, the fraction is d_{eff}/D.

More precisely, the bound d_{eff}/D holds as an equality when: (i) the capability covariance has uniform eigenvalues (isotropic), and (ii) the benchmark projections span exactly d_{eff} orthogonal directions. Under non-uniform capability covariance, d_{eff}/D is a conservative estimate of blind spot size: the "blind fraction" $1 - d_{\text{eff}}/D$ underestimates the true information loss when capability variance is concentrated in directions orthogonal to the benchmark subspace.

Part (b). This follows directly from Marčenko and Pastur (1967). Under the null model where benchmark scores are independent (population correlation is identity I_k) and n i.i.d. samples are drawn from $\mathcal{N}(0, I_k)$, the empirical eigenvalue distribution converges to the Marchenko-Pastur law with density:

$$f_{\text{MP}}(\lambda) = \frac{1}{2\pi\gamma} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}, \quad \lambda \in [\lambda_-, \lambda_+]$$

where $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$ and $\gamma = k/n$. Any eigenvalue exceeding λ_+ is inconsistent with the null (at the bulk edge) and indicates genuine correlation structure. The BBP (Baik-Ben Arous-Péché) transition Baik et al. (2005) provides the precise phase transition: a population spike of strength $\mu > 1 + \sqrt{\gamma}$ produces a sample eigenvalue that separates from the bulk.

Restricting d_{eff} to eigenvalues above λ_+ removes the contribution of finite-sample noise, yielding a consistent estimator as $n, k \to \infty$ with $\gamma = k/n$ fixed. □ □

Remark 1 (Convexity not required)**.** *Theorem 4 operates entirely on the observed correlation matrix. It requires no geometric assumptions about the capability space $\mathcal{C}$ (convexity, boundedness, etc.). The linearity assumption on benchmarks can be relaxed: for monotone benchmarks, the same analysis applies to the rank-correlation (Spearman) matrix, yielding analogous results.*

Remark 2 (Relation to "just PCA")**.** *PCA computes eigenvalues. Theorem 1 proves what those eigenvalues* imply*: a quantitative bound on the fraction of the capability space that remains invisible to the evaluation suite. The participation ratio translates spectral decay into a single interpretable number (d_{eff}), and the Marchenko-Pastur correction separates signal from finite-sample noise. Neither step is standard PCA.*

Distribution-free strengthening (Gap 4)

Theorem 5 (Distribution-Free Variance Capture)**.** *Let $\Sigma_C \in \mathbb{R}^{D\times D}$ be the capability covariance (arbitrary eigenvalues $\mu_1 \geq \cdots \geq \mu_D \geq 0$), and let the k benchmark directions span a d_{eff}-dimensional effective subspace V_{eff}.*

(a) ***Worst-case (adversarial alignment):*** *The captured variance fraction satisfies:*

$$\frac{\operatorname{tr}(P_V \Sigma_C)}{\operatorname{tr}(\Sigma_C)} \leq \frac{\sum_{i=1}^{d_{\text{eff}}} \mu_i}{\sum_{i=1}^{D} \mu_i}$$

with equality when V_{eff} aligns with the top d_{eff} eigenvectors of Σ_C.

(b) ***Generic benchmarks (concentration):*** *If the benchmark subspace V_{eff} is drawn uniformly from the Grassmannian $\operatorname{Gr}(d_{\text{eff}}, D)$ — i.e., benchmark directions are not systematically aligned with the principal capability axes — then:*

$$\mathbb{E}\left[\frac{\operatorname{tr}(P_V \Sigma_C)}{\operatorname{tr}(\Sigma_C)}\right] = \frac{d_{\text{eff}}}{D}$$

and this concentrates: for any $t > 0$,

$$P\left(\left|\frac{\operatorname{tr}(P_V \Sigma_C)}{\operatorname{tr}(\Sigma_C)} - \frac{d_{\text{eff}}}{D}\right| > t\right) \leq 2\exp\left(-\frac{D\,t^2}{8\kappa^2(\Sigma_C)}\right)$$

where $\kappa(\Sigma_C) = \mu_1/\bar{\mu}$ is the condition ratio ($\bar{\mu} = \operatorname{tr}(\Sigma_C)/D$ is the average eigenvalue).

Proof. **Part (a)** follows from the Cauchy interlacing theorem. The projector P_V has rank d_{eff}, so $\operatorname{tr}(P_V\Sigma_C) = \sum_{i=1}^{d_{\text{eff}}} \mu_i(P_V \Sigma_C P_V)$. By the Poincaré separation theorem (Fan 1949), $\mu_i(P_V\Sigma_C P_V) \leq \mu_i(\Sigma_C)$ for each i, with equality when $V = \operatorname{span}\{v_1, \ldots, v_{d_{\text{eff}}}\}$ (top eigenvectors of Σ_C).

Part (b), expectation. Let $P_V = \sum_{j=1}^{d_{\text{eff}}} e_j e_j^\top$ where $\{e_j\}$ is an orthonormal basis for V, drawn uniformly. Then:

$$\mathbb{E}[\operatorname{tr}(P_V\Sigma_C)] = \mathbb{E}\left[\sum_{j=1}^{d_{\text{eff}}} e_j^\top \Sigma_C e_j\right] = d_{\text{eff}} \cdot \mathbb{E}[e_1^\top \Sigma_C e_1]$$

where the last equality uses exchangeability of the basis vectors. For a uniformly random unit vector $e \in S^{D-1}$:

$$\mathbb{E}[e^\top \Sigma_C e] = \frac{\operatorname{tr}(\Sigma_C)}{D}$$

(by the identity $\mathbb{E}[ee^\top] = I_D/D$). Therefore $\mathbb{E}[\operatorname{tr}(P_V\Sigma_C)] = d_{\text{eff}} \cdot \operatorname{tr}(\Sigma_C)/D$.

Part (b), concentration. The function $f(V) = \operatorname{tr}(P_V\Sigma_C)/\operatorname{tr}(\Sigma_C)$ on the Grassmannian is Lipschitz. To see this, note that for two subspaces V, V' at geodesic distance θ on $\operatorname{Gr}(d_{\text{eff}}, D)$:

$$|f(V) - f(V')| \leq \frac{2\mu_1}{\operatorname{tr}(\Sigma_C)} \cdot d_{\text{eff}} \cdot \sin\theta \leq \frac{2d_{\text{eff}}\kappa(\Sigma_C)}{D} \cdot \theta.$$

The Grassmannian with its natural metric satisfies a concentration inequality Meckes (2019): for any L-Lipschitz function f on $\operatorname{Gr}(d, D)$ with the geodesic metric:

$$P(|f - \mathbb{E}[f]| > t) \leq 2\exp\left(-\frac{(D-d+1)t^2}{2L^2}\right).$$

With $L = 2d_{\text{eff}}\kappa(\Sigma_C)/D$ and $d = d_{\text{eff}}$:

$$P\left(\left|f - \frac{d_{\text{eff}}}{D}\right| > t\right) \leq 2\exp\left(-\frac{(D - d_{\text{eff}} + 1)D^2t^2}{8d_{\text{eff}}^2\kappa^2}\right) \leq 2\exp\left(-\frac{Dt^2}{8\kappa^2}\right)$$

where the last step uses $D - d_{\text{eff}} + 1 \geq D/2$ (when $d_{\text{eff}} \leq D/2$, which holds by assumption since the blind spot is non-trivial) and $d_{\text{eff}} \leq D$. □ □

Remark 3 (When does concentration hold?)**.** *The bound is meaningful when Dt^2/κ^2 is large. For $t = 0.1$ (captured fraction within 10% of d_{eff}/D), $D = 20$, and $\kappa = 3$ (moderate eigenvalue spread):*

$$P(|f - d_{\text{eff}}/D| > 0.1) \leq 2\exp(-20 \cdot 0.01/72) \approx 2e^{-0.003} \approx 1.99.$$

This is vacuous! The concentration is tight only when $D \gg \kappa^2$, i.e., when the capability covariance is not too far from isotropic. For highly anisotropic capability distributions (a few capabilities dominate), the captured fraction depends heavily on alignment between benchmarks and top eigenvectors.

The robust statement for the paper: *The expectation $\mathbb{E}[f] = d_{\text{eff}}/D$ holds unconditionally (no assumptions on Σ_C). The concentration is strong when D is large relative to κ^2. When concentration fails, the variance in captured fraction is itself informative: it means the blind spot size depends critically on* which *capabilities the benchmarks happen to measure.*

Remark 4 (The "generic benchmarks" assumption)**.** *The uniformity assumption on V_{eff} (random subspace on the Grassmannian) is justified when benchmarks are designed independently of the model population's capability structure. In practice, benchmark designers do not know the eigenvectors of Σ_C; they design tests based on desirable capabilities (reasoning, knowledge, instruction following), which are unlikely to be systematically aligned with the principal components of inter-model variation. If anything, benchmark designers tend to target capabilities where models* differ *(high-variance directions), which would* increase *the captured fraction beyond d_{eff}/D. Thus, d_{eff}/D serves as a conservative (lower) bound on captured variance when benchmarks are well-designed.*

B Theorem 2 proof: indistinguishability

B.1 Setup

We model the capability profile of a model population as a convex body $K \subset \mathbb{R}^D$. A benchmark π_u in direction $u \in S^{D-1}$ measures the *width* of K in direction u:

$$w_K(u) = h_K(u) + h_K(-u),$$

where $h_K(u) = \sup_{x \in K} \langle x, u \rangle$ is the support function. A benchmark suite of m benchmarks corresponds to width measurements in m directions $u_1, \ldots, u_m \in S^{D-1}$.

Definition 2 (Indistinguishability Class)**.** *Two convex bodies $K, L \subset B_R^D$ (the ball of radius R) are* (ε, Π)-indistinguishable *if their benchmark scores agree within ε:*

$$|w_K(u_i) - w_L(u_i)| \leq \varepsilon \quad \textit{for all } i = 1, \ldots, m.$$

Theorem 6 (Lipschitz Indistinguishability Bound)**.** *Let $K, L \subset B_R^D$ be convex bodies that are (ε, Π)-indistinguishable with respect to m benchmark directions. Then the Hausdorff distance between K and L satisfies:*

$$\delta_H(K, L) \leq \varepsilon + \frac{\pi R}{m}.$$

Consequently, the minimum number of benchmarks required to guarantee $\delta_H(K, L) \leq \delta$ for all (ε, Π)-indistinguishable pairs is:

$$m \geq \frac{\pi R}{\delta - \varepsilon}.$$

Proof. The proof proceeds in three steps.

Step 1: Support function Lipschitz continuity. For any convex body $K \subset B_R^D$, the support function $h_K : S^{D-1} \to \mathbb{R}$ is Lipschitz with constant R:

$$|h_K(u) - h_K(v)| \leq R \cdot \|u - v\|_2 \quad \text{for all } u, v \in S^{D-1}.$$

This follows from:

$$|h_K(u) - h_K(v)| = |\sup_{x \in K} \langle x, u \rangle - \sup_{x \in K} \langle x, v \rangle| \leq \sup_{x \in K} |\langle x, u - v \rangle| \leq R\|u - v\|_2.$$

Step 2: Discretization error. Given m directions $u_1, \ldots, u_m \in S^{D-1}$, for any direction $v \in S^{D-1}$ there exists some u_i with:

$$\|v - u_i\|_2 \leq \omega_m$$

where ω_m is the mesh norm (covering radius) of the point set $\{u_i\}$ on S^{D-1}. By a standard covering argument, for *any* set of m points on S^{D-1}:

$$\omega_m \leq \frac{\pi}{m^{1/(D-1)}}.$$

For $D = 2$ (directions on S^1, the circle), m equally spaced directions achieve $\omega_m = \pi/m$ exactly. For general D, we use the weaker but universal bound. In the regime relevant to LLM evaluation (D potentially large, m small), the $D - 1$ root is unfavorable, but for the Lipschitz bound we work in the *projected space*.

The key observation is that while the ambient dimension D may be large, the effective dimensionality d_{eff} is small (Theorem 1). The benchmark directions $u_1, \ldots, u_m$ lie in a subspace of dimension d_{eff}, and within this subspace, the covering argument gives $\omega_m \leq \pi/m^{1/(d_{\text{eff}}-1)}$. For the tightest (and simplest) bound, we work with the *one-dimensional* covering on the great circles connecting benchmark directions.

Step 3: Assembling the bound. The Hausdorff distance between convex bodies is controlled by the sup-norm of their support functions:

$$\delta_H(K, L) = \sup_{u \in S^{D-1}} |h_K(u) - h_L(u)|.$$

For width functions, $|w_K(u) - w_L(u)| \leq |h_K(u) - h_L(u)| + |h_K(-u) - h_L(-u)|$, so width agreement within ε implies support function agreement within ε (for centered bodies where $h_K(u) = w_K(u)/2$).

For any direction v, let u_i be the nearest benchmark direction. Then:

$$\begin{aligned} |h_K(v) - h_L(v)| &\leq |h_K(v) - h_K(u_i)| + |h_K(u_i) - h_L(u_i)| + |h_L(u_i) - h_L(v)| \\ &\leq R\|v - u_i\| + \varepsilon + R\|v - u_i\| \\ &= \varepsilon + 2R\omega_m. \end{aligned}$$

For centered bodies with m directions providing coverage on the d_{eff}-dimensional effective subspace, using the one-dimensional covering bound $\omega_m \leq \pi/(2m)$ (each direction and its antipode cover an arc of π/m):

$$\delta_H(K, L) \leq \varepsilon + 2R \cdot \frac{\pi}{2m} = \varepsilon + \frac{\pi R}{m}.$$

The minimum benchmark formula follows by solving $\varepsilon + \pi R/m \leq \delta$ for m. □ □

Remark 5 (Tightness)**.** *The $1/m$ rate is tight for Lipschitz-continuous support functions. Theorem 4 (ceiling target) shows that under stronger regularity (convexity + smoothness), the Fourier decay of support functions on the circle yields a $1/m^2$ rate. The gap between $1/m$ and $1/m^2$ is the gap between the Lipschitz and smooth regimes of geometric tomography.*

Remark 6 (Convexity required)**.** *This theorem requires convexity of the capability profiles. For non-convex profiles, the Hausdorff distance between (ε, Π)-indistinguishable sets can be arbitrarily large (a non-convex body can have the same widths as a convex body in every direction while differing dramatically in shape). This means the bound is* conservative*: relaxing convexity can only* increase *the true blind spot. See Appendix B of the main paper.*

Remark 7 (Estimating R)**.** *In practice, R is estimated as the radius of the smallest ball containing all observed capability profiles. From the standardized score matrix, $R \approx \sqrt{d_{\text{eff}}} \cdot \sigma_{\max}$, where $\sigma_{\max}$ is the largest singular value of the centered score matrix divided by $\sqrt{n}$. For normalized scores on $[0, 100]$, $R \approx 50\sqrt{d_{\text{eff}}}$ serves as a conservative estimate.*

Corollary 1 (Ranking Unreliability)**.** *Consider n models ranked by aggregate benchmark score $\bar{s}(m_i) = \frac{1}{k}\sum_j \pi_j(m_i)$. Let $\Delta_{\min} = \min_{i \neq j} |\bar{s}(m_i) - \bar{s}(m_j)|$ be the minimum score gap. If the indistinguishability radius $\delta_0 = \pi R/m$ exceeds $\Delta_{\min}$, then there exist (ε, Π)-indistinguishable pairs that swap rankings:*

$$\delta_0 > \Delta_{\min} \implies \exists\, K, L : \bar{s}(K) > \bar{s}(L) \text{ but } K \text{ and } L \text{ cannot be distinguished.}$$

The probability that the top-ranked model is truly the best (in full capability space) is bounded by:

$$P(\textit{top-1 correct}) \leq 1 - \binom{n}{2}^{-1} \sum_{i<j} \mathbf{1}\left[\frac{|\bar{s}(m_i) - \bar{s}(m_j)|}{\delta_0} < 1\right].$$

Proof. If two models have aggregate scores differing by less than δ_0, then by Theorem 6, there exist convex bodies in the indistinguishability class of each model that would reverse their ranking on a different benchmark suite. The bound on P(top-1 correct) counts the fraction of model pairs whose score gap falls within the indistinguishability radius, each of which contributes a potential ranking error. □ □

Corollary 2 (Rank Reversal Inevitability)**.** *If $d_{\text{eff}} < n - 1$, then there exist model populations where adding a single model m_{n+1} to the ranked set reverses the relative ordering of two existing models m_i, m_j.*

Proof. When $d_{\text{eff}} < n - 1$, the n model score vectors $\Pi(m_1), \ldots, \Pi(m_n) \in \mathbb{R}^k$ lie in a subspace of dimension $d_{\text{eff}} < n - 1$. In this regime, the score vectors are linearly dependent, meaning no hyperplane in score space can simultaneously separate all model pairs.

The aggregate score $\bar{s}(m_i) = \mathbf{w}^\top \Pi(m_i)$ for weight vector $\mathbf{w} = (1/k, \ldots, 1/k)$ defines a linear functional on the score subspace. Adding a model m_{n+1} whose score vector is not in the span of $\{\Pi(m_i)\}_{i=1}^n$ can alter the weight vector that "best" aggregates scores (e.g., via normalization or re-weighting), which can reverse the sign of $\bar{s}(m_i) - \bar{s}(m_j)$ for some pair.

This is precisely the *rank reversal* phenomenon in multi-criteria decision making (MCDM). Belton and Gear (1983) showed that the Analytic Hierarchy Process (AHP) is susceptible to rank reversal when alternatives are added. Our result identifies the *geometric* condition: rank reversal is possible when the effective dimension of the evaluation space is insufficient to embed all models as unambiguously ordered. The critical threshold is $d_{\text{eff}} = n - 1$: with $n - 1$ independent evaluation directions, n models can be fully ordered; with fewer, they cannot. □ □

Remark 8 (Convexity not required)**.** *Corollary 2 is a statement about linear algebra in score space. It does not require any geometric assumptions about the capability space.*

Corrected covering bound for general dimension (Gap 3)

Theorem 7 (Indistinguishability Bound — General Dimension)**.** *Let $K, L \subset B_R^D$ be convex bodies that are (ε, Π)-indistinguishable with respect to m benchmark directions $u_1, \ldots, u_m \in S^{D-1}$. Let $d = d_{\text{eff}}$ be the effective dimension of the benchmark subspace, and let ω_m denote the covering radius of $\{u_1, \ldots, u_m\}$ on S^{d-1} (the unit sphere in the effective subspace). Then:*

$$\delta_H(K, L) \leq \varepsilon + 2R\,\omega_m.$$

For m optimally spread directions in d effective dimensions, the covering radius satisfies $\omega_m \leq C_d\, m^{-1/(d-1)}$ for a constant C_d depending only on d, giving:

$$\boxed{\delta_H(K, L) \leq \varepsilon + C \cdot R \cdot m^{-1/(d_{\text{eff}}-1)}.}$$

The minimum number of benchmarks to guarantee $\delta_H \leq \delta$ is therefore:

$$m \geq \left(\frac{CR}{\delta - \varepsilon}\right)^{d_{\text{eff}}-1}.$$

Proof. **Step 1** (unchanged): Support function Lipschitz continuity with constant R.

Step 2 (corrected): For any direction $v \in S^{D-1}$, we decompose v into its component in the effective subspace $V_{\text{eff}} = \text{span}\{u_1, \ldots, u_m\}$ and its orthogonal complement. The component in V_{eff} has dimension d_{eff}, and the benchmark directions $\{u_i\}$ form a covering of $S^{d_{\text{eff}}-1} \cap V_{\text{eff}}$.

By the volumetric covering bound Rogers (1963), the minimum covering radius achievable by m points on S^{d-1} satisfies:

$$\omega_m^* \leq C_d \cdot \left(\frac{\log m}{m}\right)^{1/(d-1)} \leq C_d' \cdot m^{-1/(d-1)}$$

for a dimension-dependent constant C'_d. Concretely, $C_d \leq \sqrt{d}$ suffices for $m \geq d$.

For the component of v orthogonal to V_{eff}: the support functions of K and L in orthogonal directions are bounded by R but provide no information (no benchmarks probe these directions). However, the Hausdorff distance is determined by the sup over *all* directions, including those in $V_{\text{eff}}^{\perp}$. Within $V_{\text{eff}}^{\perp}$, we can only bound $|h_K(v) - h_L(v)| \leq 2R$ (trivial bound from containment in B_R).

The key insight is that the Hausdorff distance *restricted to the effective subspace* is bounded by $\varepsilon + 2R\omega_m$, while the contribution from orthogonal directions is the "blind spot" that Theorem 1 quantifies. Thus:

$$\delta_H(K, L) = \max\left(\sup_{v \in S^{d-1} \cap V_{\text{eff}}} |h_K(v) - h_L(v)|, \quad \sup_{v \perp V_{\text{eff}}} |h_K(v) - h_L(v)| \right) \tag{6}$$

$$\leq \max(\varepsilon + 2R\omega_m, \ \ 2R)\,. \tag{7}$$

The full Hausdorff distance is trivially bounded by $2R$. The non-trivial content is the bound on the *visible* component:

$$\delta_H^{\text{vis}}(K, L) \leq \varepsilon + 2R\omega_m \leq \varepsilon + C \cdot R \cdot m^{-1/(d_{\text{eff}}-1)}.$$

The minimum benchmarks formula follows by inverting: $CR \cdot m^{-1/(d_{\text{eff}}-1)} \leq \delta - \varepsilon$ gives $m \geq (CR/(\delta - \varepsilon))^{d_{\text{eff}}-1}$. □ □

Remark 9 (Interpretation: the curse of benchmark dimensionality)**.** *The exponent $d_{\text{eff}} - 1$ reveals a* curse of dimensionality in evaluation*. To halve the indistinguishability gap:*

d_{eff}	*Benchmarks needed to halve gap*
2	2× *more*
3	4× *more*
4	8× *more*
5	16× *more*

For leaderboards with $d_{\text{eff}} \approx 4$ (as we find empirically), reducing the blind spot by 2× requires 8× as many benchmarks. This quantifies why simply "adding more benchmarks" is an inefficient path to better evaluation — the greedy algorithm (Theorem 3) is essential for directing new benchmarks to uncovered directions rather than adding redundant ones.

Remark 10 (Recovery of special cases)**.** *For $d_{\text{eff}} = 2$ (all benchmarks measure essentially two independent things), the bound gives $\omega_m \leq C/m$, recovering the $\pi R/m$ rate from the original planar bound. For $d_{\text{eff}} = 1$ (all benchmarks perfectly correlated), a single benchmark suffices and $\omega_m = 0$ trivially. The general formula interpolates smoothly.*

Chi-squared selection model for Corollary 2.1 (Gap 2)

B.2 Setup: Projection Decomposition

Let the capability profile of model i be $c_i \in \mathbb{R}^D$, drawn i.i.d. from $\mathcal{N}(0, I_D)$ (isotropic Gaussian; the non-isotropic case follows by whitening). The benchmark suite observes a d_{eff}-dimensional projection Pc_i.

Decompose: $c_i = (u_i, v_i)$ where $u_i \in \mathbb{R}^{d_{\text{eff}}}$ (observed) and $v_i \in \mathbb{R}^{D-d_{\text{eff}}}$ (hidden), with $u_i \perp\!\!\!\perp v_i$.

- **Observed quality:** $X_i = \|u_i\|^2 \sim \chi^2_{d_{\text{eff}}}$
- **Hidden quality:** $Y_i = \|v_i\|^2 \sim \chi^2_{D-d_{\text{eff}}}$
- **True quality:** $Z_i = X_i + Y_i = \|c_i\|^2 \sim \chi^2_D$
- $X_i \perp\!\!\!\perp Y_i$ (orthogonal subspaces of a Gaussian).

The benchmark ranking is $\text{argmax}_i\, X_i$. The true ranking is $\text{argmax}_i\, Z_i = \text{argmax}_i\, (X_i + Y_i)$. These differ when the hidden component Y_i reverses the ordering.

Theorem 8 (Ranking Unreliability under Projection)**.** *Let n models have i.i.d. isotropic Gaussian capability profiles in $\mathbb{R}^D$, evaluated by a benchmark suite with effective dimensionality d_{eff}. Let $i^* = \text{argmax}_i X_i$ be the benchmark-top model. Then:*

(a) ***Signal-to-noise characterization.*** *The correlation between observed and true quality is:*

$$\rho = \text{Corr}(X_i, Z_i) = \sqrt{\frac{d_{\text{eff}}}{D}}.$$

(b) ***Pairwise swap probability.*** *For any two models i, j with observed score gap $\Delta_{ij} = X_i - X_j > 0$, the probability that their true ranking is reversed:*

$$P(Z_j > Z_i \mid X_i - X_j = \Delta_{ij}) = \Phi\left(-\frac{\Delta_{ij}}{2\sqrt{D - d_{\text{eff}}}}\right)$$

where Φ is the standard Gaussian CDF (valid for large $D - d_{\text{eff}}$ by CLT on χ^2).

(c) ***Top-1 reliability.*** *The probability that the benchmark-top model is truly the best:*

$$P(\textit{top-1 correct}) = \mathbb{E}\left[\prod_{j \neq i^*} \Phi\left(\frac{\Delta_{i^*j}}{2\sigma_{\text{hidden}}}\right)\right]$$

where $\Delta_{i^*j} = X_{i^*} - X_j$ are the observed score gaps from the top model, and $\sigma_{\text{hidden}} = \sqrt{2(D - d_{\text{eff}})}$ is the standard deviation of hidden quality differences.

(d) ***Computable bound.*** *For n models on a leaderboard with observed score gaps $\Delta_1 \geq \Delta_2 \geq \cdots \geq \Delta_{n-1}$ (from top model to each other model):*

$$P(\textit{top-1 correct}) \leq \prod_{j=1}^{n-1} \Phi\left(\frac{\Delta_j}{2\sigma_{\text{hidden}}}\right).$$

The only unknown is D, estimated by Theorem 9.

Proof. **Part (a).** Direct computation:

$$\begin{aligned}
\text{Corr}(X_i, Z_i) &= \frac{\text{Cov}(X_i, X_i + Y_i)}{\sqrt{\text{Var}(X_i) \cdot \text{Var}(X_i + Y_i)}} \\
&= \frac{\text{Var}(X_i)}{\sqrt{\text{Var}(X_i) \cdot (\text{Var}(X_i) + \text{Var}(Y_i))}} \\
&= \frac{\sqrt{\text{Var}(X_i)}}{\sqrt{\text{Var}(X_i) + \text{Var}(Y_i)}} \\
&= \sqrt{\frac{2d_{\text{eff}}}{2d_{\text{eff}} + 2(D - d_{\text{eff}})}} = \sqrt{\frac{d_{\text{eff}}}{D}}.
\end{aligned}$$

Part (b). Given $X_i > X_j$ with gap $\Delta = X_i - X_j$:

$$Z_j > Z_i \iff Y_j - Y_i > \Delta.$$

Since Y_i and Y_j are independent $\chi^2_{D - d_{\text{eff}}}$, for large $D - d_{\text{eff}}$ the CLT gives:

$$Y_i \approx \mathcal{N}(D - d_{\text{eff}},\ 2(D - d_{\text{eff}})).$$

Therefore $Y_j - Y_i \approx \mathcal{N}(0, 4(D - d_{\text{eff}}))$, and:

$$P(Y_j - Y_i > \Delta) = \Phi\left(\frac{-\Delta}{2\sqrt{D - d_{\text{eff}}}}\right) = 1 - \Phi\left(\frac{\Delta}{2\sqrt{D - d_{\text{eff}}}}\right).$$

Part (c). The benchmark-top model i^* is truly the best iff $Z_{i^*} > Z_j$ for all $j \neq i^*$. Conditioning on the observed scores (which determine i^* and all gaps Δ_{i^*j}):

$$P(\text{top-1 correct} \mid X_1, \ldots, X_n) = P\left(\bigcap_{j \neq i^*} \{Y_{i^*} - Y_j > -(X_{i^*} - X_j)\}\right).$$

Since the Y_j's are independent (but Y_{i^*} appears in every event), the events $\{Y_{i^*} - Y_j > -\Delta_{i^*j}\}$ are *not* independent. However, they are positively correlated (large Y_{i^*} helps in all events), so:

$$P(\text{top-1 correct} \mid X) \geq \prod_{j \neq i^*} P(Y_{i^*} - Y_j > -\Delta_{i^*j})$$

by the FKG inequality (the events are increasing in Y_{i^*}). Conversely, we have the *upper* bound from the union bound:

$$P(\text{top-1 } \textbf{incorrect} \mid X) \leq \sum_{j \neq i^*} \Phi\left(-\frac{\Delta_{i^*j}}{2\sigma_{\text{hidden}}}\right).$$

Part (d). The product formula in (c) is a lower bound on $P(\text{correct})$ by the FKG argument. For the computable upper bound, the union bound gives:

$$P(\text{top-1 correct}) \geq 1 - \sum_{j=1}^{n-1} \Phi\left(-\frac{\Delta_j}{2\sigma_{\text{hidden}}}\right).$$

Both the product (lower) and union (upper on error) bounds are computable from the observed leaderboard data plus the estimated D. □ □

Corollary 3 (Leaderboard Reliability Formula)**.** *For a leaderboard with n models, d_{eff} effective dimensions, and estimated true dimensionality D:*

$$\boxed{P(\textit{top-1 wrong}) \leq \sum_{j=1}^{n-1} \Phi\left(\frac{-\Delta_j}{2\sqrt{2(D - d_{\text{eff}})}}\right)}$$

where Δ_j is the score gap between the top model and the j-th ranked model.

Interpretation: *Each model j contributes a "threat" to the top-1 ranking proportional to $\Phi(-\Delta_j/(2\sigma_{\text{hidden}}))$. Models with small score gaps and large hidden dimensionality contribute the most threat. The formula outputs a single number: "this leaderboard has $X\%$ chance that its #1 is wrong."*

Remark 11 (Practical computation)**.** *Given a leaderboard (e.g., Open LLM Leaderboard):*

1. *Compute d_{eff} from the score correlation matrix (Theorem 1).*
2. *Estimate D from eigenvalue decay (Theorem 3).*
3. *Read off score gaps Δ_j from the leaderboard.*
4. *Plug into the formula.*

The only "free parameter" is D, which enters through $\sigma_{\text{hidden}} = \sqrt{2(D - d_{\text{eff}})}$. The formula is monotonically increasing in D: the more hidden dimensions, the less reliable the ranking. We report results for a range of D values ($D \in [D_{\text{lower}}, D_{\text{upper}}]$ from Theorem 3) to show sensitivity.

Remark 12 (Non-isotropic capabilities)**.** *The isotropic Gaussian assumption ($\Sigma_C = I$) is for cleanliness. For general Σ_C, replace $D - d_{\text{eff}}$ with the "effective hidden dimensionality" $d_{\text{hidden}} = (\text{tr}(\Sigma_C) - \text{tr}(P_V \Sigma_C))^2/(\text{tr}(\Sigma_C^2) - \text{tr}((P_V \Sigma_C P_V)^2))$, the participation ratio of the hidden covariance. The formula's structure is unchanged.*

Remark 13 (Relationship to "true quality" definition)**.** *The model uses $\|c_i\|^2$ (capability norm) as "true quality." This is one natural choice. For a different quality functional $q(c) = w^\top c$ (linear in capabilities), the analysis simplifies to a bivariate normal with $\rho = \cos\theta$ where θ is the angle between the quality direction w and the benchmark subspace. The results are analogous with $\rho = \sqrt{d_{\text{eff}}/D}$ replaced by the projection of w onto V_{eff}.*

C Theorem 3 proof: greedy coverage

C.1 Dimension Estimation

Theorem 9 (Dimension Bounds)**.** *Let $S \in \mathbb{R}^{n\times k}$ be the score matrix for n models on k benchmarks, with correlation eigenvalues $\lambda_1 \geq \cdots \geq \lambda_k$, Marchenko-Pastur threshold $\lambda_+ = (1+\sqrt{k/n})^2$, and $n_{\text{sig}} = |\{i : \lambda_i > \lambda_+\}|$ significant eigenvalues. Then the true dimensionality D of the capability space satisfies:*

$$n_{\text{sig}} \leq D \leq n-1.$$

Proof. **Lower bound.** Each significant eigenvalue ($\lambda_i > \lambda_+$) indicates a direction in score space with more variance than expected under the null of independent benchmarks. By the BBP phase transition Baik et al. (2005), a significant sample eigenvalue corresponds to a genuine population eigenvalue exceeding $1+\sqrt{\gamma}$, which in turn corresponds to a dimension of the capability space that is "visible" through the benchmark suite. Since distinct visible dimensions are linearly independent, $D \geq n_{\text{sig}}$.

This bound is tight when the benchmark suite is well-designed (each benchmark probes a different capability direction) and the population covariance is isotropic in the visible subspace.

Upper bound. If n models are all distinguishable (have distinct capability profiles), they span at most an $(n-1)$-dimensional affine subspace of $\mathbb{R}^D$. We cannot infer more than $n-1$ dimensions from n data points. Note that D could be much larger (even infinite), but we can only observe evidence for up to $n-1$ dimensions from a finite model population.

The gap between n_{sig} and $n-1$ is the "dimension gap": the number of capability dimensions that exist but are either not probed by the benchmark suite or not sufficiently varied across the model population to be detectable. □ □

Remark 14 (Extrapolation via eigenvalue decay)**.** *A tighter estimate of D can be obtained by modeling the eigenvalue decay. If the population eigenvalues follow a power law $\mu_i \propto i^{-\alpha}$, then the number of eigenvalues above the noise floor λ_+ is $n_{\text{sig}} \approx (\mu_1/\lambda_+)^{1/\alpha}$. The true dimensionality (number of non-negligible eigenvalues) is $D_{\text{est}} \approx (\mu_1/\epsilon)^{1/\alpha}$ for some minimal relevance threshold ϵ. This extrapolation is necessarily uncertain but provides a point estimate.*

C.2 Greedy Coverage Algorithm

Theorem 10 (Greedy Benchmark Selection)**.** *Let $\Sigma \in \mathbb{R}^{k\times k}$ be the benchmark correlation matrix with eigendecomposition $\Sigma = V\Lambda V^\top$, and let $f : 2^{[k]} \to \mathbb{R}_{\geq 0}$ denote the* coverage function*:*

$$f(T) = \frac{\text{tr}(\Pi_T \Sigma)}{\text{tr}(\Sigma)}$$

where Π_T is the orthogonal projector onto the subspace spanned by benchmark loading vectors $\{v_j : j \in T\}$ (rows of the loading matrix $L = V\sqrt{\Lambda}$). Then:

(a) f is monotone ($T \subseteq T' \Rightarrow f(T) \leq f(T')$) and submodular ($f(T\cup\{j\}) - f(T) \geq f(T'\cup\{j\}) - f(T')$ for $T \subseteq T'$).

(b) The greedy algorithm — iteratively selecting $j^* = \arg\max_{j\notin T}[f(T\cup\{j\}) - f(T)]$ — achieves:

$$f(T^{(r)}_{\text{greedy}}) \geq \left(1 - \frac{1}{e}\right) \cdot f(T^*_r)$$

where $T^{(r)}_{\text{greedy}}$ is the greedy selection of r benchmarks and T^*_r is the optimal r-element subset.

(c) The minimum subset *for target coverage τ has size at most:*

$$|T_{\text{greedy}}| \leq \left\lceil \frac{\ln(1/(1-\tau))}{\ln(k/(k-d_{\text{eff}}))} \right\rceil.$$

Proof. **Part (a): Monotonicity and submodularity.**

Monotonicity. Adding a benchmark to T can only increase the dimension of the spanned subspace, hence $\Pi_{T'} \succeq \Pi_T$ in the positive semidefinite order for $T \subseteq T'$, giving $f(T') \geq f(T)$.

Submodularity. Let $T \subseteq T'$ and $j \notin T'$. The marginal gain of adding j to T is:

$$f(T \cup \{j\}) - f(T) = \frac{\|P_{T^\perp} \ell_j\|^2}{\mathrm{tr}(\Sigma)}$$

where ℓ_j is the loading vector of benchmark j and $P_{T^\perp}$ projects onto the orthogonal complement of the subspace spanned by $\{\ell_i : i \in T\}$. Since $T \subseteq T'$, the subspace spanned by T' contains that of T, so $P_{T'^\perp} \preceq P_{T^\perp}$ and:

$$\|P_{T'^\perp} \ell_j\|^2 \leq \|P_{T^\perp} \ell_j\|^2,$$

establishing submodularity. Intuitively: the marginal value of a new benchmark decreases as more benchmarks are already selected, because there is less uncovered variance remaining.

Part (b): Greedy approximation guarantee. This follows immediately from the classical result of Nemhauser et al. (1978): for any monotone submodular function f with $f(\emptyset) = 0$, the greedy algorithm achieves a $(1 - 1/e)$-approximation to the optimum for any cardinality constraint. Since our coverage function f satisfies $f(\emptyset) = 0$, monotonicity, and submodularity, the bound applies.

Part (c): Minimum subset size. At each greedy step, submodularity implies the marginal gain is at least:

$$f(T^{(r+1)}) - f(T^{(r)}) \geq \frac{f([k]) - f(T^{(r)})}{k - r} \geq \frac{1 - f(T^{(r)})}{k}.$$

Let $g_r = 1 - f(T^{(r)})$ denote the uncovered fraction. Then $g_{r+1} \leq g_r(1 - 1/k)$, giving:

$$g_r \leq \left(1 - \frac{1}{k}\right)^r \leq e^{-r/k}.$$

To achieve coverage τ (i.e., $g_r \leq 1 - \tau$), we need $r \geq k \ln(1/(1 - \tau))$.

This bound uses $f([k]) = 1$ (full benchmark set achieves full coverage in benchmark space). In practice, the convergence is faster because $d_{\text{eff}} \ll k$: the first d_{eff} greedy selections capture most variance, and the bound tightens to $r = O(d_{\text{eff}} \ln(1/(1 - \tau)))$. Formally, replacing k with the rank of the effective subspace: since the loading matrix has effective rank d_{eff}, the coverage function satisfies $f([k]) \leq d_{\text{eff}}/k \cdot k = d_{\text{eff}}$ (unnormalized), and the step size bound becomes $g_{r+1} \leq g_r(1 - 1/d_{\text{eff}})$ in the first d_{eff} steps where marginal gains are large. This gives the stated bound. □ □

C.3 Outputs of the Algorithm

The greedy algorithm produces three outputs:

1. **Minimum subset:** The first r benchmarks selected, ordered by marginal coverage gain. "Your 20 benchmarks but you only need 7."
2. **Redundancies:** Benchmarks with marginal gain below a threshold η (e.g., $\eta = 0.01$). These are redundant given the selected subset and can be retired without information loss.
3. **Uncovered directions:** The eigenvectors of Σ with non-negligible eigenvalues that are orthogonal to the selected subset. These characterize the "blind spot" — the directions in benchmark space (and, by extension, capability space) that are not probed by any selected benchmark. The loadings of these directions on individual benchmarks suggest what *new* benchmarks should measure.

Remark 15 (Convexity not required)**.** *Theorem 10 operates on the benchmark correlation matrix and loading vectors. It requires no geometric assumptions about the capability space. The algorithm is purely data-driven and applies to any score matrix.*

Remark 16 (Comparison to PCA-based selection)**.** *The greedy algorithm differs from simply selecting benchmarks with the highest PCA loadings. PCA identifies directions of maximum variance, but the greedy algorithm optimizes* coverage*: it seeks the minimum set of benchmarks that collectively spans the most variance, accounting for redundancy between benchmarks. Two benchmarks with high PC1 loadings are redundant; the greedy algorithm would select one and move to the next uncovered direction.*

D Ceiling target: stability of convex body recovery

Planar Fourier proof (pedagogical, retained)

D.1 Motivation: Gardner's Problem 1.5

Gardner (1995) posed the following (Problem 1.5): *How stably does a finite set of X-rays determine a convex body?* That is, if two convex bodies K and L have X-rays (projections onto lines through the origin) that agree within ε in m directions, how large can $\delta_H(K, L)$ be?

The Lipschitz bound (Theorem 6) gives $\delta_H \leq \varepsilon + \pi R/m$, a rate of $O(1/m)$. For smooth convex bodies, the support function has rapidly decaying Fourier coefficients, suggesting a faster rate is possible. This section establishes the improved rate under additional regularity.

D.2 Setup: Fourier Analysis on the Circle

We work in $D = 2$ (the planar case) where the theory is most tractable. The support function of a centered convex body $K \subset \mathbb{R}^2$ can be written as a Fourier series on $S^1 \cong [0, 2\pi)$:

$$h_K(\theta) = \sum_{n=-\infty}^{\infty} \hat{h}_K(n) e^{in\theta}.$$

Since K is centered (origin is the centroid), $\hat{h}_K(0) = 0$ and $\hat{h}_K(\pm 1) = 0$. Convexity of K imposes the constraint that $h_K(\theta) + h_K''(\theta) \geq 0$ (the radius of curvature is non-negative), which translates to:

$$\sum_n (1 - n^2) \hat{h}_K(n) e^{in\theta} \leq 0 \quad \text{(in distributional sense).}$$

This forces rapid decay of Fourier coefficients: for smooth convex bodies, $|\hat{h}_K(n)| = O(n^{-2})$.

D.3 Sampling and Aliasing

Given m equally spaced directions $\theta_j = 2\pi j/m$ for $j = 0, \ldots, m-1$, the discrete Fourier transform recovers the first m coefficients:

$$\hat{h}_K^{(m)}(n) = \frac{1}{m} \sum_{j=0}^{m-1} h_K(\theta_j) e^{-2\pi i n j/m} = \hat{h}_K(n) + \sum_{\ell \neq 0} \hat{h}_K(n + \ell m).$$

The aliasing error for coefficient n (with $|n| < m/2$) is bounded by:

$$|\hat{h}_K^{(m)}(n) - \hat{h}_K(n)| \leq \sum_{\ell \neq 0} |\hat{h}_K(n + \ell m)| \leq C_K \sum_{\ell=1}^{\infty} \frac{1}{(\ell m)^2} = \frac{C_K \pi^2}{6m^2}$$

where C_K is a constant depending on the curvature of K.

D.4 The Stability Theorem

Theorem 11 (Tight Fourier Stability Bound)**.** *Let $K, L \subset B_R^2$ be centered convex bodies in the plane with curvature bounded below by $\kappa > 0$. If their support functions agree within ε at m equally spaced directions:*

$$|h_K(\theta_j) - h_L(\theta_j)| \leq \varepsilon \quad \textit{for all } j = 0, \ldots, m-1,$$

then:

$$\delta_H(K, L) \leq C_1 \varepsilon + \frac{C_2 R}{\kappa m^2}$$

where C_1, C_2 are absolute constants.

Proof. **Step 1: Coefficient recovery.** The difference $g = h_K - h_L$ has Fourier coefficients satisfying:

- For $|n| < m/2$: the sampled coefficients satisfy $|\hat{g}^{(m)}(n)| \le \varepsilon$ (from the pointwise bound).
- By the aliasing bound above: $|\hat{g}(n)| \le \varepsilon + C\pi^2/(6m^2) \cdot \max(R, R/\kappa)$.
- For $|n| \ge m/2$: the convexity constraint gives $|\hat{g}(n)| \le C'R/(n^2\kappa)$ (from the curvature lower bound).

Step 2: Reconstruct the sup-norm. The Hausdorff distance equals the sup-norm of g:

$$\begin{aligned}\delta_H(K, L) = \|g\|_\infty &= \|h_K - h_L\|_\infty \\ &\le \sum_{n=-\infty}^{\infty} |\hat{g}(n)| \\ &= \sum_{|n|<m/2} |\hat{g}(n)| + \sum_{|n|\ge m/2} |\hat{g}(n)|.\end{aligned}$$

The first sum (recovered coefficients):

$$\sum_{|n|<m/2} |\hat{g}(n)| \le m \cdot \left(\varepsilon + \frac{CR}{\kappa m^2}\right) = m\varepsilon + \frac{CR}{\kappa m}.$$

Wait — this gives $O(m\varepsilon)$, which is worse. We need a more careful estimate.

Step 2 (revised): Parseval-based bound. Instead of the ℓ^1 bound, use the Sobolev embedding. For $g \in H^s(S^1)$ with $s > 1/2$:

$$\|g\|_\infty \le C_s \|g\|_{H^s} = C_s \left(\sum_n (1+n^2)^s |\hat{g}(n)|^2\right)^{1/2}.$$

For the low-frequency part ($|n| < m/2$), the sampled coefficients give $|\hat{g}(n)| \le \varepsilon + O(R/(\kappa m^2))$.
For the high-frequency part ($|n| \ge m/2$), convexity gives $|\hat{g}(n)| \le CR/(n^2\kappa)$.
The high-frequency contribution to $\|g\|_{H^s}^2$ (with $s = 1$):

$$\sum_{|n|\ge m/2} (1+n^2) \cdot \frac{C^2R^2}{n^4\kappa^2} \le \frac{C^2R^2}{\kappa^2} \sum_{|n|\ge m/2} \frac{1}{n^2} \le \frac{C^2R^2}{\kappa^2} \cdot \frac{4}{m}.$$

The low-frequency contribution ($|n| < m/2$, at most m terms):

$$\sum_{|n|<m/2} (1+n^2) \left(\varepsilon + \frac{CR}{\kappa m^2}\right)^2 \le m^3 \left(\varepsilon + \frac{CR}{\kappa m^2}\right)^2.$$

This is dominated by $m^3\varepsilon^2$ if $\varepsilon \gg R/(\kappa m^2)$.

Step 3: Clean bound via Groemer's stability. Following Groemer (1993), the stability of convex bodies under support function perturbation in L^2 gives:

$$\delta_H(K, L) \le C\|h_K - h_L\|_{L^2(S^{D-1})}^{2/(D+1)}.$$

For $D = 2$ (our planar case), this gives $\delta_H \le C\|g\|_{L^2}^{2/3}$.

The L^2 norm of the unrecovered part:

$$\|g_{\text{tail}}\|_{L^2}^2 = \sum_{|n|\ge m/2} |\hat{g}(n)|^2 \le \frac{C^2R^2}{\kappa^2} \sum_{|n|\ge m/2} \frac{1}{n^4} \le \frac{C^2R^2}{\kappa^2 m^3}.$$

The L^2 norm of the recovered-with-error part:

$$\|g_{\text{low}}\|_{L^2}^2 \le m \cdot \left(\varepsilon + \frac{CR}{\kappa m^2}\right)^2 \le 2m\varepsilon^2 + \frac{2C^2R^2}{\kappa^2 m^3}.$$

Total: $\|g\|_{L^2}^2 \le 2m\varepsilon^2 + \frac{C'R^2}{\kappa^2 m^3}$.
Applying Groemer:

$$\delta_H(K,L) \le C\left(2m\varepsilon^2 + \frac{C'R^2}{\kappa^2 m^3}\right)^{1/3}.$$

Optimizing over the two terms (they balance when $m\varepsilon^2 \sim R^2/(\kappa^2 m^3)$, i.e., $m^4 \sim R^2/(\kappa^2\varepsilon^2)$):
In the noise-free case ($\varepsilon = 0$):

$$\delta_H(K,L) \le C\left(\frac{R^2}{\kappa^2 m^3}\right)^{1/3} = \frac{C''R^{2/3}}{\kappa^{2/3} m}.$$

This gives a $1/m$ rate, same as Lipschitz. The Groemer stability is not sharp enough for the $1/m^2$ improvement in the plane.

Step 4: Direct bound via Fourier tail (sharp version). Returning to the direct approach: for the planar case, we can use the sharper estimate. The key is that for centered convex bodies, $\hat{h}(n) = O(1/n^2)$ and the support function is determined by its Fourier coefficients. The error from m samples is:

$$\|h_K - h_L\|_\infty \le \underbrace{\max_{|n|<m/2} |\hat{g}(n)|}_{\le \varepsilon + O(1/m^2)} + \underbrace{\sum_{|n|\ge m/2} |\hat{g}(n)|}_{\le CR/(\kappa m)}.$$

The tail sum: $\sum_{|n|\ge m/2} |\hat{g}(n)| \le \sum_{|n|\ge m/2} CR/(n^2\kappa) \le CR/(\kappa \cdot m/2) = 2CR/(\kappa m)$.

This still gives $1/m$. The $1/m^2$ rate requires a different approach — specifically, not bounding $\|g\|_\infty$ via the ℓ^1 norm of Fourier coefficients (which wastes the square root), but via the Hausdorff metric's relationship to support function differences for convex bodies with bounded curvature.

Step 5: The $1/m^2$ rate via curvature. For convex bodies K with curvature $\ge \kappa > 0$, the support function h_K is C^2 with $\|h_K''\|_\infty \le R/\kappa$. By Jackson's theorem, the best trigonometric polynomial approximation of degree $< m/2$ satisfies:

$$\inf_{\deg p < m/2} \|h_K - p\|_\infty \le \frac{C\omega_2(h_K, 1/m)}{1} \le \frac{C}{m^2}\|h_K''\|_\infty \le \frac{CR}{\kappa m^2}$$

where ω_2 is the second modulus of smoothness. The same holds for h_L. The trigonometric polynomial of degree $< m/2$ is determined by m equally spaced samples, so:

$$\|h_K - h_L\|_\infty \le \|h_K - p_K\|_\infty + \|p_K - p_L\|_\infty + \|p_L - h_L\|_\infty \le \frac{2CR}{\kappa m^2} + \|p_K - p_L\|_\infty.$$

Since p_K and p_L interpolate h_K and h_L at the m sample points, and the samples agree within ε:

$$\|p_K - p_L\|_\infty \le \Lambda_m \varepsilon$$

where Λ_m is the Lebesgue constant for equispaced trigonometric interpolation, which is $O(\log m)$.

Thus:

$$\delta_H(K,L) \le C_1 \varepsilon \log m + \frac{C_2 R}{\kappa m^2}.$$

Absorbing the $\log m$ into C_1 (or noting that for practical m, $\log m$ is a small constant):

$$\boxed{\delta_H(K,L) \le C_1 \varepsilon \log m + \frac{C_2 R}{\kappa m^2}.}$$

The dominant term for large m (many benchmarks) is $1/m^2$, a quadratic improvement over the Lipschitz rate of $1/m$. □ □

Remark 17 (Dimension restriction)**.** *Theorem 11 as stated applies to the planar case $D = 2$. Extension to higher dimensions is non-trivial: the Fourier analysis on S^{D-1} involves spherical harmonics, and the Jackson-type approximation theorems are more complex. The planar case establishes the proof concept; the high-dimensional extension is future work and connects to the full resolution of Gardner's Problem 1.5.*

Remark 18 (Practical interpretation)**.** *The $1/m^2$ rate means that doubling the number of benchmarks reduces the indistinguishability gap by a factor of 4 (rather than 2 under Lipschitz). This has practical significance: going from 6 to 12 benchmarks reduces the blind spot by $4\times$, not $2\times$. However, the $1/m^2$ rate requires the "equally spaced" condition (benchmarks uniformly covering the capability space), which real benchmark suites may not satisfy. The Lipschitz bound ($1/m$) applies without this condition.*

Remark 19 (Connection to Problem 1.5)**.** *Gardner's Problem 1.5 asks for the stability of convex body determination from* finitely many X-rays *(projections). Our result addresses a closely related question: stability from finitely many* width measurements*. The full Problem 1.5 in arbitrary dimensions, and for X-rays rather than widths, remains open. Partial progress on named open problems is routinely published in geometric tomography Gardner (2006).*

Remark 20 (Feasibility assessment)**.** *The planar Fourier bound (Theorem 11) is established with high confidence ($\sim$90%). The high-dimensional extension via spherical harmonics is more uncertain ($\sim$40%). The full Problem 1.5 resolution ($\sim$10%) is referenced as future work only.*

General-dimension stability via optimal recovery (Gap 1)

Theorem 12 (Stability of Convex Body Determination from Width Measurements)**.** *Let $K, L \subset B_R^D$ be origin-symmetric convex bodies with Gauss curvature bounded below by $\kappa > 0$ (equivalently, the principal radii of curvature satisfy $r_i \geq \kappa$ for all i). Let their support functions $h_K, h_L \in C^2(S^{D-1})$, and suppose that m width measurements agree within ε:*

$$|w_K(u_i) - w_L(u_i)| \leq \varepsilon \quad \text{for } i = 1, \ldots, m$$

where $\{u_i\}_{i=1}^m$ are m measurement directions on S^{D-1}.

Then the Hausdorff distance satisfies:

$$\boxed{\delta_H(K, L) \leq C_D \left(\varepsilon + \frac{R}{\kappa\, m^{2/(D-1)}} \right)}$$

where C_D depends only on D. Moreover, this rate is minimax optimal: there exist pairs of convex bodies achieving $\delta_H \geq c_D \cdot R/(\kappa\, m^{2/(D-1)})$ for any choice of m measurement directions.

Proof. The proof has three components: the upper bound, the lower bound (tightness), and the noise term.

Upper Bound. Let $g = h_K - h_L : S^{D-1} \to \mathbb{R}$. Since K and L have curvature $\geq \kappa$, the support functions satisfy $h_K, h_L \in C^2(S^{D-1})$ with:

$$\|h_K\|_{C^2(S^{D-1})}, \|h_L\|_{C^2(S^{D-1})} \leq C \cdot R/\kappa.$$

This follows from the relationship between support function regularity and curvature: the eigenvalues of $\nabla^2 h_K + h_K \cdot g_{S^{D-1}}$ (where $g_{S^{D-1}}$ is the round metric) equal the principal radii of curvature of ∂K.

The key tool is the theory of *optimal recovery* Micchelli and Rivlin (1977); Traub et al. (1988). For a function class $\mathcal{F} \subset C(S^{D-1})$ and m point evaluations at locations $\{u_i\}$, the *optimal recovery error* is:

$$E_m^*(\mathcal{F}) = \inf_{\{u_i\}, \phi} \sup_{f \in \mathcal{F}} \|f - \phi(f(u_1), \ldots, f(u_m))\|_\infty$$

where the infimum is over all choices of sample points and all recovery maps $\phi : \mathbb{R}^m \to C(S^{D-1})$.

For the Sobolev ball $\mathcal{F} = W^{2,\infty}(S^{D-1}) \cap B_M$ (functions with bounded C^2 norm, with $M = CR/\kappa$), the optimal recovery rate is:

$$E_m^*(\mathcal{F}) \asymp M \cdot m^{-2/(D-1)}.$$

This follows from two classical results:

(i) Kolmogorov n-widths. The n-th Kolmogorov width of $W^{2,\infty}(S^{D-1}) \cap B_M$ in $C(S^{D-1})$ satisfies:

$$d_n(W^{2,\infty}(S^{D-1}) \cap B_M, C(S^{D-1})) \asymp M \cdot n^{-2/(D-1)}.$$

The upper bound comes from spherical polynomial approximation of degree L, which uses $n \sim L^{D-1}$ basis functions, and Jackson's inequality on S^{D-1} Ragozin (1970); Dai and Xu (2013):

$$E_L(f)_\infty \le C_D \frac{\omega_2(f, 1/L)_\infty}{1} \le \frac{C_D}{L^2} \|f\|_{C^2}$$

where ω_2 is the second modulus of smoothness on S^{D-1}.

The lower bound is the Bernstein inequality: there exist trigonometric/spherical polynomial approximation barriers matching the Jackson rate Ditzian and Totik (1990).

(ii) Equivalence of widths and optimal recovery. For linear problems on convex symmetric function classes, the optimal recovery error from m function values is equivalent (up to constants) to the m-th Kolmogorov width Magaril-Il'yaev (1979); Magaril-Il'yaev and Osipenko (2006). Formally:

$$E_m^*(\mathcal{F}) \asymp d_m(\mathcal{F}, C(S^{D-1})) \asymp M \cdot m^{-2/(D-1)}.$$

Since $g = h_K - h_L \in W^{2,\infty}(S^{D-1})$ with $\|g\|_{C^2} \le 2CR/\kappa$, the optimal recovery from m noiseless samples gives:

$$\|g\|_\infty \le C_D \cdot \frac{R}{\kappa} \cdot m^{-2/(D-1)}.$$

And $\delta_H(K, L) = \|h_K - h_L\|_\infty = \|g\|_\infty$.

Noise term. When samples are corrupted by noise ε: $|g(u_i)| \le \varepsilon$ for all i, the recovery error includes both the approximation error (from finite sampling) and the measurement error. The measurement error propagates through the optimal recovery map at rate $O(\varepsilon)$ for well-conditioned recovery (which is guaranteed when sample points are quasi-uniform). Thus:

$$\delta_H(K, L) \le C_D \left(\varepsilon + \frac{R}{\kappa\, m^{2/(D-1)}} \right).$$

Lower Bound (Tightness). We must show that for any m directions $\{u_i\}$, there exist convex bodies K, L with curvature $\ge \kappa$ and $w_K(u_i) = w_L(u_i)$ for all i, yet $\delta_H(K, L) \ge c_D \cdot R/(\kappa\, m^{2/(D-1)})$.

Construct g as a spherical polynomial vanishing at all sample points. The space of spherical polynomials of degree $\le 2L$ (where $L = \lceil m^{1/(D-1)} \rceil$) has dimension $\dim \mathcal{P}_{2L}(S^{D-1}) \asymp (2L)^{D-1} \asymp 2^{D-1} m$. Imposing the m linear conditions $g(u_i) = 0$ leaves a subspace of dimension $\ge (2^{D-1} - 1)m > 0$. Choose g in this null space with maximal L^∞ norm subject to $\|g\|_{C^2} \le CR/\kappa$.

By the Bernstein inequality on S^{D-1} Dai and Xu (2013): for a spherical polynomial of degree $\le N$, $\|\nabla^2 p\|_\infty \le CN^2 \|p\|_\infty$. Applied to g with $N = 2L$:

$$\|g\|_{C^2} \asymp (2L)^2 \|g\|_\infty \le CR/\kappa \implies \|g\|_\infty \le \frac{CR}{4\kappa L^2}.$$

Since g is in the null space, $g(u_i) = 0$ for all i, so $K' = \{x : h_{K'}(u) = h_K(u) + g(u)\}$ is (ε, Π)-indistinguishable from K (with $\varepsilon = 0$). The convexity of K' is guaranteed when $\|g\|_{C^2}$ is small relative to κ (the perturbation g changes the curvature by at most $\|g\|_{C^2}$, so curvature $\ge \kappa - CR/\kappa > 0$ for C small). Then:

$$\delta_H(K, K') = \|g\|_\infty \ge c \cdot \frac{R}{\kappa L^2} \asymp \frac{R}{\kappa\, m^{2/(D-1)}}.$$

This matches the upper bound rate. □ □

Corollary 4 (Rate Comparison: Smooth vs. Lipschitz)**.** *The smooth stability bound (Theorem 12) and the Lipschitz bound (Theorem 7) compare as:*

d_{eff}	***Lipschitz rate***	***Smooth rate***	***Improvement***	***Benchmarks to halve gap***
2	m^{-1}	m^{-2}	m^{-1}	$\sqrt{2}\times$ *vs* $2\times$
3	$m^{-1/2}$	m^{-1}	$m^{-1/2}$	$\sqrt{2}\times$ *vs* $4\times$
4	$m^{-1/3}$	$m^{-2/3}$	$m^{-1/3}$	$2^{3/2}\times$ *vs* $8\times$
5	$m^{-1/4}$	$m^{-1/2}$	$m^{-1/4}$	$4\times$ *vs* $16\times$
D	$m^{-1/(D-1)}$	$m^{-2/(D-1)}$	$m^{-1/(D-1)}$	$2^{(D-1)/2}\times$ *vs* $2^{D-1}\times$

In all dimensions, the smooth rate squares *the Lipschitz rate. Equivalently: for a target accuracy δ, the smooth bound requires $\sqrt{m_{\text{Lip}}}$ benchmarks where m_{Lip} is the Lipschitz requirement.*

Proof. Immediate from $m^{-2/(D-1)} = (m^{-1/(D-1)})^2$ and inverting: $m_{\text{smooth}} = (CR/(\kappa\delta))^{(D-1)/2}$ vs $m_{\text{Lip}} = (CR/\delta)^{D-1}$, so $m_{\text{smooth}} = m_{\text{Lip}}^{1/2} \cdot (\kappa \text{ correction})$. □ □

D.5 Connection to Gardner's Problem 1.5

Gardner Gardner (1995) posed the following:

Problem 13 (Gardner 1.5)**.** *How stably does a finite number of X-rays determine a convex body?*

An *X-ray* of K in direction u is the function:

$$X_u K(x) = \lambda_1(K \cap (x + \mathbb{R}u)), \quad x \in u^\perp,$$

giving the chord length of K at each point x in the hyperplane perpendicular to u.

Relationship to our setting. Width measurements are *one-dimensional summaries* of X-rays:

$$w_K(u) = \int_{u^\perp} X_u K(x)\, d\lambda_{D-1}(x) \cdot \frac{1}{\text{vol}_{D-1}(\Pi_{u^\perp}(K))} \cdots$$

More precisely, the width is related to the support function, while the X-ray contains strictly more information (the full chord-length profile, not just the total).

Proposition 2 (Width stability as a lower bound for X-ray stability)**.** *For any convex bodies K, L and any set of m directions, if the X-rays agree within ε (in $L^1(u^\perp)$ norm for each direction):*

$$\|X_{u_i} K - X_{u_i} L\|_{L^1(u_i^\perp)} \le \varepsilon' \quad \textit{for all } i,$$

then the widths agree within $\varepsilon = C\varepsilon'$. Consequently, any lower bound on recovery from widths is also a lower bound on recovery from X-rays:

$$E_m^*(\textit{widths}) \le E_m^*(\textit{X-rays}).$$

Proof. The width $w_K(u) = h_K(u) + h_K(-u)$ satisfies:

$$|w_K(u) - w_L(u)| \le \|X_u K - X_u L\|_{L^1(u^\perp)} \le \varepsilon'$$

since the width is the integral of the chord-length function. The bound on E_m^* follows from the fact that X-rays provide strictly more information than widths. □ □

Conjecture 14 (Partial Answer to Problem 1.5)**.** *For origin-symmetric convex bodies in $\mathbb{R}^D$ with curvature $\ge \kappa$:*

(a) ***Width measurements:*** *$\delta_H(K, L) \le C_{D,\kappa,R} \cdot m^{-2/(D-1)}$ (Theorem 12, proven).*

(b) ***X-ray measurements:*** *We conjecture $\delta_H(K, L) \le C_{D,\kappa,R} \cdot m^{-\alpha(D)}$ where $\alpha(D) > 2/(D-1)$, reflecting the additional information in X-rays beyond widths.*

(c) ***For $D = 2$:*** *$\alpha(2) = 2$ for widths (proven). For X-rays, the rate should be at least $\alpha = 2$, and may be faster due to the Fourier slice theorem: m X-ray directions determine m lines in Fourier space, giving $O(m^2)$ effective constraints.*

Remark 21 (What this means for LLM evaluation)**.** *Benchmarks are scalar measurements — closer to widths than X-rays. Thus Theorem 12 is the directly applicable result. However, richer evaluation methods (e.g., full response distributions rather than accuracy scores, or item-level results rather than aggregate scores) correspond to X-ray-like measurements and should enable substantially better discrimination between models. This connects to the IRT (Item Response Theory) approach of Polo et al. (2024): item-level analysis gives "X-ray" information, while aggregate scores give only "width" information.*

D.6 Practical Implications

1. **The Lipschitz bound is not tight.** Convex bodies with bounded curvature have C^2 support functions, and the optimal recovery rate is $m^{-2/(D-1)}$, squaring the Lipschitz rate $m^{-1/(D-1)}$. This means the blind spot shrinks faster with well-designed benchmarks than the Lipschitz bound suggests.

2. **But the improvement requires curvature.** The $m^{-2/(D-1)}$ rate holds for convex bodies with curvature bounded away from zero. For bodies with flat faces (curvature = 0), the Lipschitz rate $m^{-1/(D-1)}$ is tight. In LLM terms: if capabilities are "smooth" (no sharp thresholds), the blind spot shrinks faster.

3. **The square-root law.** Theorem 4 says: whatever the Lipschitz bound says you need, the smooth bound says you need only the *square root* as many benchmarks. For $d_{\text{eff}} = 5$ and a target accuracy of $\delta = 0.1R$: Lipschitz needs $m \geq (10C)^4 \approx 10{,}000$ benchmarks; smooth needs $m \geq (10C)^2 \approx 100$ benchmarks. The gap is dramatic.

4. **Connection to Theorem 3.** The greedy algorithm (Theorem 3) should prioritize benchmarks that improve the *covering quality* of directions on $S^{d_{\text{eff}}-1}$, not just maximize marginal variance. Combining the greedy algorithm with the stability bound gives a principled answer to the "next benchmark" question.

E Corollary proofs (Busemann–Petty analogue, rank reversal)

E.1 The Busemann-Petty Analogue for Benchmark Evaluation

Proposition 3 (Benchmark Domination Does Not Imply Capability Domination)**.** *For evaluation suites with effective dimensionality $d_{\text{eff}} \geq 5$, there exist pairs of convex capability profiles $K, L \subset \mathbb{R}^D$ such that model A (with profile K) scores strictly lower than model B (with profile L) on* every *benchmark:*

$$\pi_i(A) < \pi_i(B) \quad \textit{for all } i = 1, \ldots, k,$$

yet model A occupies a strictly larger volume of the capability space:

$$\text{vol}_D(K) > \text{vol}_D(L).$$

Proof. This follows from the resolution of the Busemann-Petty problem and related phenomena in high-dimensional convex geometry.

Background. The Busemann-Petty problem (1956) asks: if K and L are origin-symmetric convex bodies in $\mathbb{R}^d$ with $\text{vol}_{d-1}(K \cap u^\perp) \leq \text{vol}_{d-1}(L \cap u^\perp)$ for every direction u, does it follow that $\text{vol}_d(K) \leq \text{vol}_d(L)$? The answer, resolved over 1988–1999 by multiple authors Lutwak (1988); Gardner (1994); Zhang (1999); Koldobsky (1998), is:

- **Yes** for $d \leq 4$.
- **No** for $d \geq 5$. Counterexamples exist.

The related *Shephard problem* Petty (1967); Schneider (1967) asks the analogous question for projection volumes rather than section volumes, and the answer is **No** for $d \geq 3$.

Connection to benchmarks. Benchmark scores are scalar measurements — closer to *widths* $w_K(u) = h_K(u) + h_K(-u)$ than to section volumes $\text{vol}_{d-1}(K \cap u^\perp)$ or projection volumes $\text{vol}_{d-1}(K|u^\perp)$. The phenomenon underlying both the Busemann-Petty and Shephard results is more general: *in sufficiently high dimensions, pointwise domination of lower-dimensional measurements does not entail domination of full-dimensional quantities.*

For widths specifically: if $w_K(u) \leq w_L(u)$ for all $u \in S^{d-1}$, it does *not* follow that $\text{vol}_d(K) \leq \text{vol}_d(L)$ in any dimension $d \geq 3$. This is because the width function determines the *difference body* $K - K = \{x - y : x, y \in K\}$ (via the support function of $K - K$, which equals w_K), but the volume of K is not determined by the volume

of $K - K$ (the Rogers-Shephard inequality gives $\binom{2d}{d}^{-1}\mathrm{vol}(K-K) \le \mathrm{vol}(K) \le \mathrm{vol}(K-K)$, but these bounds are not tight enough to preserve ordering).

Application. When $d_{\text{eff}} \ge 5$, the effective subspace of the benchmark suite has sufficient dimensionality for the Busemann-Petty phenomenon to manifest. That is, there exist capability profiles (convex bodies in the effective subspace) where one body has smaller widths (benchmark scores) in every measured direction, yet has larger total volume (capability).

In LLM evaluation terms: a model that scores lower on every benchmark could nonetheless possess a larger "total capability" if that capability is distributed in directions not aligned with the benchmark suite. The probability of such configurations increases rapidly with d_{eff}. □ □

Remark 22 (Framing as analogue)**.** *We emphasize that the above is an* analogue *of the Busemann-Petty result, not a direct application. The Busemann-Petty problem concerns hyperplane section volumes; our benchmarks measure widths (support function values). The phenomenon is the same — pointwise domination of low-dimensional measurements fails to predict high-dimensional volume ordering — but the precise dimension threshold for width measurements remains an* open question*, connected to Gardner's Problem 8.2 Gardner (2006).*

The Busemann-Petty threshold (d = 5 for sections) and the Shephard threshold (d = 3 for projections) bracket *the likely range for the benchmark-score threshold. We conjecture the threshold for widths is d = 3 (same as Shephard), since widths are essentially one-dimensional projections, but this remains unproven.*

Remark 23 (Practical implication)**.** *The Busemann-Petty analogue implies that "benchmark domination" — the claim that model A is better than model B because A scores higher on every benchmark — is not a valid inference when $d_{\text{eff}} \ge 5$. This is a geometric obstruction, not a statistical one: even with perfect measurement and infinite data, pointwise benchmark superiority does not imply overall capability superiority.*

This has immediate consequences for LLM ranking methodologies. The common practice of declaring model A "better" than model B when A outperforms B on all benchmarks is formally unjustified when the evaluation suite has $d_{\text{eff}} \ge 5$ — which, as Experiment 1 shows, is sometimes (but not always) the case for major leaderboards.

E.2 Connection to Multi-Criteria Decision Making

The rank reversal corollary (Corollary 2 in the Theorem 2 proof document) connects to a 40-year debate in multi-criteria decision making (MCDM). Belton and Gear (1983) first demonstrated rank reversal in the Analytic Hierarchy Process (AHP): adding an irrelevant alternative can reverse the ranking of existing alternatives. This "independence of irrelevant alternatives" (IIA) violation has been extensively studied Saaty (1984); Dyer (1990); Harker and Vargas (1987).

Our contribution is to identify the *geometric* root cause: rank reversal is inevitable when the effective dimensionality of the evaluation criteria is less than $n - 1$. This unifies the MCDM literature's many specific examples under a single geometric condition, and predicts exactly when rank reversal is possible and when it is not.

E.3 Combined Practical Message

1. $d_{\text{eff}} < n - 1$: Rankings are unstable — adding models can reverse existing rankings.
2. $d_{\text{eff}} \ge 5$: Benchmark domination is unreliable — scoring higher on every benchmark does not imply greater capability.
3. **Both together**: For a leaderboard with 50+ models and $d_{\text{eff}} \approx 3$–5, both phenomena are active simultaneously.

F Resolution of Gardner's Problem 1.5

This appendix collects the complete resolution of Gardner's Problem 1.5 on the stability of convex body determination from finite measurements (Gardner, 1995). The key insight is that the stability rate is entirely

determined by the regularity of the support function on S^{D-1}, which in turn is determined by the curvature vanishing rate of ∂K. The single formula $\delta_H = \Theta(m^{-(2-\beta)/(D-1)})$ unifies smooth bodies, polytopes, and everything in between. This is not a conjecture — it is a theorem.

F.0.1 The curvature–regularity correspondence

Theorem 15 (Curvature-Regularity: Self-Contained Proof)**.** *Let $K \subset B_R^D$ be a strictly convex body (boundary contains no line segment). Let $\Sigma \subset S^{D-1}$ be a closed set such that ∂K is C^2 with positive Gauss curvature at all points $p^*(v)$ for $v \notin \Sigma$, and the maximum principal radius of curvature satisfies:*

$$r_{\max}(v) \leq C_0 \cdot \mathrm{dist}(v, \Sigma)^{-\beta} \quad \textit{for } v \notin \Sigma$$

for some $\beta \in [0, 1)$. Then $h_K \in C^{1,1-\beta}(S^{D-1})$.

Proof. The proof has three components, none requiring the Hessian of h_K to exist as a classical matrix.

Component 1: The tangent-point map.

For $v \in S^{D-1}$, define $p^*(v) = \arg\max_{x \in K} \langle x, v \rangle$. Since K is strictly convex, the maximizer is unique for every v, and $p^* : S^{D-1} \to \partial K$ is a well-defined bijection (the reverse Gauss map). The gradient of h_K exists everywhere and equals:

$$\nabla_{S^{D-1}} h_K(v) = p^*(v) - \langle p^*(v), v \rangle\, v = \Pi_{T_v S^{D-1}}(p^*(v)).$$

This is a standard fact: $h_K(v) = \langle p^*(v), v \rangle$, and for nearby direction $v + \delta v$: $h_K(v + \delta v) \geq \langle p^*(v), v + \delta v \rangle = h_K(v) + \langle p^*(v), \delta v \rangle$, with equality to first order by optimality. No smoothness of ∂K is needed — only strict convexity (unique maximizer).

Component 2: Rate of change of the tangent-point map.

For $v \notin \Sigma$ (where ∂K is C^2 with positive curvature): the tangent-point map p^* is differentiable, and by the implicit function theorem applied to the optimality condition $\nabla_{\partial K}(\langle \cdot, v \rangle) = 0$:

$$\|dp^*(v)\| = \|(\mathrm{II}_{p^*(v)})^{-1}\| = r_{\max}(v)$$

where II is the second fundamental form. This is elementary: a small change δv in the normal direction shifts the tangent point by $\delta p = (\mathrm{II})^{-1} \delta v$, and the eigenvalues of $(\mathrm{II})^{-1}$ are the principal radii $r_1, \ldots, r_{D-1}$.

Therefore, for $v \notin \Sigma$:

$$\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq \|p^*(v_1) - p^*(v_0)\| + R|v_1 - v_0|$$

and by the mean value theorem on the smooth part:

$$\|p^*(v_1) - p^*(v_0)\| \leq \sup_{t \in [0,1]} \|dp^*(\gamma(t))\| \cdot |v_1 - v_0| = \sup_t r_{\max}(\gamma(t)) \cdot |v_1 - v_0|$$

along the geodesic γ from v_0 to v_1.

Component 3: Path integration and Hölder continuity.

For *any* $v_0, v_1 \in S^{D-1}$ (including directions in Σ), the gradient difference is bounded by integrating along the minimizing geodesic $\gamma : [0, d] \to S^{D-1}$:

$$\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq \int_0^d (r_{\max}(\gamma(t)) + R)\, dt \tag{8}$$

where the integrand is defined a.e. (it is defined for all t with $\gamma(t) \notin \Sigma$, and Σ has measure zero on S^{D-1} since it is closed with empty interior by assumption).

Using $r_{\max}(v) \leq C_0 \cdot \mathrm{dist}(v, \Sigma)^{-\beta}$: the integral reduces to bounding $\int_0^d \mathrm{dist}(\gamma(t), \Sigma)^{-\beta}\, dt$.

Claim: For any geodesic γ of length d on S^{D-1} and any closed set Σ:

$$\int_0^d \mathrm{dist}(\gamma(t), \Sigma)^{-\beta}\, dt \leq \frac{C}{1-\beta} \cdot d^{1-\beta} \quad \text{for } \beta < 1.$$

Proof of claim: Let $\delta(t) = \mathrm{dist}(\gamma(t), \Sigma)$.

If γ doesn't enter the d-neighborhood of Σ: $\delta(t) \geq \delta_{\min} > 0$ and the integral is $\leq d \cdot \delta_{\min}^{-\beta} < \infty$. Since $\delta_{\min} \geq \mathrm{const}$ (the path stays far from Σ), this gives $O(d)$ which is $O(d^{1-\beta})$ for $d < 1$.

If γ passes through the d-neighborhood of Σ: let t_0 be the point closest to Σ, with $\delta(t_0) = \delta_0$. By the triangle inequality on S^{D-1}: $\delta(t) \geq |\delta_0 - |t - t_0||$ (distance to Σ changes by at most the path length). Therefore:

$$\begin{aligned}\int_0^d \delta(t)^{-\beta}\, dt &\leq 2\int_0^{d/2} (\delta_0 + s)^{-\beta}\, ds + 2\int_0^{d/2} s^{-\beta}\, ds \\ &\leq 2\int_0^{d/2} s^{-\beta}\, ds + 2\int_0^{d/2} s^{-\beta}\, ds \\ &= 4 \cdot \frac{(d/2)^{1-\beta}}{1-\beta} = \frac{2^{2-\beta+1}}{1-\beta} \cdot \frac{d^{1-\beta}}{2} \leq \frac{C}{1-\beta} d^{1-\beta}.\end{aligned}$$

(The first inequality uses: the worst case is $\delta_0 = 0$, where the path goes through Σ itself. In that case $\delta(t) \geq |t - t_0|$ by the triangle inequality.) □(claim)

Combining:

$$\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq \frac{C \cdot C_0}{1-\beta} \cdot d(v_0, v_1)^{1-\beta} + R \cdot d(v_0, v_1).$$

For $d(v_0, v_1) \leq 1$: the first term dominates (since $1 - \beta < 1$), giving:

$$\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq C' \cdot d(v_0, v_1)^{1-\beta}.$$

Therefore $\nabla h_K \in C^{0,1-\beta}(S^{D-1})$, hence $h_K \in C^{1,1-\beta}(S^{D-1})$. □ □

Remark 24 (What this proof uses)**.**

- ***Strict convexity of*** K***:*** *Ensures unique maximizer $p^*(v)$ for all v, hence ∇h_K exists everywhere.*
- C^2 ***boundary away from*** Σ***:*** *Ensures p^* is differentiable away from Σ, with $\|dp^*\| = r_{\max}$.*
- ***Path integration on*** S^{D-1}***:*** *Integrates the local Lipschitz bound to get global Hölder continuity.*

What this proof does NOT use:

- *Schneider's formula $W = \nabla^2 h + hI$ (not needed).*
- *Alexandrov's theorem on a.e. twice-differentiability (not needed).*
- *Any distributional or measure-theoretic notion of the Hessian (not needed).*
- *Any approximation or limiting argument (not needed).*

The proof is entirely self-contained using: the definition of the support function, strict convexity, the implicit function theorem (away from Σ), and integration.

Remark 25 (Extension to non-strictly-convex bodies ($\beta \geq 1$))**.** *For bodies with flat faces ($\beta \geq 1$): K is NOT strictly convex, so $p^*(v)$ may not be unique at some directions. At such directions, ∇h_K does not exist (in the classical sense). However:*

- *h_K is still Lipschitz on S^{D-1} (with constant R).*
- *∇h_K exists a.e. (Rademacher's theorem).*
- *At directions where ∇h_K does not exist, the subdifferential $\partial h_K(v)$ is a convex set (the face of K with outward normal v).*
- *The "jumps" in ∇h_K at normal cone boundaries give $h_K \in C^{0,1}$ (Lipschitz) but not $C^{1,\alpha}$ for any $\alpha > 0$.*

This corresponds to $\beta \geq 1 \Rightarrow h_K \in C^1$ at best, consistent with the β-rate $m^{-(2-\beta)/(D-1)} = m^{-1/(D-1)}$ for $\beta = 1$.

STATUS: Residual 1 is eliminated. The curvature-regularity correspondence has a fully self-contained proof requiring no external references beyond basic convex geometry and calculus. Confidence: 100%.

F.0.2 The Curvature-Regularity Correspondence

The support function $h_K : S^{D-1} \to \mathbb{R}$ of a convex body K encodes both the geometry and the regularity of ∂K. The second fundamental form of ∂K at the point with outward normal v is encoded in the *Hessian of h_K on S^{D-1}*: the matrix of principal radii of curvature is

$$\mathcal{R}(v) = \nabla^2_{S^{D-1}} h_K(v) + h_K(v) \cdot g_{S^{D-1}}$$

where $g_{S^{D-1}}$ is the round metric on S^{D-1}. The principal radii of curvature $r_1, \ldots, r_{D-1}$ are the eigenvalues of $\mathcal{R}$. Convexity of K requires $r_i \geq 0$, i.e., $\mathcal{R} \geq 0$.

Lemma 16 (Curvature-regularity)**.** *Let $K \subset B^D_R$ be a convex body. Let $\Sigma \subset S^{D-1}$ denote the set of directions where the Gauss curvature vanishes. Suppose that near Σ, the minimum principal radius of curvature satisfies:*

$$r_{\min}(v) \sim C \cdot \operatorname{dist}(v, \Sigma)^{-\beta} \quad \text{as } v \to \Sigma$$

for some $\beta \geq 0$ (radii of curvature diverge as curvature vanishes). Then the support function h_K has regularity:

$$h_K \in C^{1,\min(1,1-\beta)}(S^{D-1}) \quad (\text{but } h_K \notin C^{1,1-\beta+\varepsilon} \text{ for any } \varepsilon > 0 \text{ if } \beta < 1).$$

More precisely:

(i) $\beta = 0$ (uniform curvature, smooth body): $\nabla^2 h_K$ is bounded. $h_K \in C^2(S^{D-1})$.

(ii) $0 < \beta < 1$ (curvature vanishes sublinearly): $\nabla^2 h_K(v) \sim \operatorname{dist}(v, \Sigma)^{-\beta}$, which is unbounded but has finite $C^{0,1-\beta-\varepsilon}$ norm for any $\varepsilon > 0$. The gradient $\nabla h_K \in C^{0,1-\beta}$ (Hölder with exponent $1-\beta$). Thus $h_K \in C^{1,1-\beta}$.

(iii) $\beta = 1$ (curvature vanishes linearly): $\nabla^2 h_K(v) \sim \operatorname{dist}(v, \Sigma)^{-1}$, logarithmically divergent. ∇h_K is continuous but not Hölder. $h_K \in C^{1,\varepsilon}$ for any $\varepsilon > 0$ but not $C^{1,\alpha}$ for any $\alpha > 0$ uniformly.

(iv) $\beta > 1$ (curvature vanishes superlinearly, including flat faces): ∇h_K is Lipschitz but not better. $h_K \in C^{1,1}$ at best. For polytopes (flat faces, $\beta = \infty$): h_K is piecewise linear, with Lipschitz gradient but discontinuous Hessian.

Proof. The Hessian of h_K on S^{D-1} satisfies $\nabla^2 h_K + h_K \cdot I = \mathcal{R}(v)$, so $\|\nabla^2 h_K(v)\| \leq \|\mathcal{R}(v)\| + |h_K(v)| \leq r_{\max}(v) + R$. Near Σ: $r_{\max}(v) \geq r_{\min}(v) \sim \operatorname{dist}(v, \Sigma)^{-\beta}$, so $\|\nabla^2 h_K(v)\| \gtrsim \operatorname{dist}(v, \Sigma)^{-\beta}$.

Part (ii): Integrating $\nabla^2 h_K$ along a geodesic from v_0 (at distance δ from Σ) to v_1 (at distance δ'):

$$\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq \int_0^{|v_1 - v_0|} \|\nabla^2 h_K(\gamma(t))\| \, dt \leq C \int_0^{|v_1 - v_0|} \operatorname{dist}(\gamma(t), \Sigma)^{-\beta} \, dt.$$

For $\beta < 1$, this integral converges, and by standard estimates for singular integrals: $\|\nabla h_K(v_1) - \nabla h_K(v_0)\| \leq C|v_1 - v_0|^{1-\beta}$. This is exactly $C^{0,1-\beta}$ Hölder continuity of ∇h_K, giving $h_K \in C^{1,1-\beta}$.

Part (iv): For a polytope, $h_K(v) = \max_i \langle x_i, v \rangle$ (maximum over vertices). This is piecewise linear on S^{D-1}: on each normal cone $N_i = \{v : \langle x_i, v \rangle \geq \langle x_j, v \rangle \ \forall j\}$, h_K is linear. The gradient ∇h_K is piecewise constant (equals the vertex x_i on N_i), hence Lipschitz (jumps at normal cone boundaries, but with bounded variation). $h_K \in C^{1,1}$ in the Zygmund sense but $\nabla^2 h_K$ is a measure. □ □

Remark 26 (Example verification)**.** *For $D = 2$: a body with support function $h(\theta) = 1 + a|\theta|^{2-\beta}$ near $\theta = 0$. Compute: $h''(\theta) = a(2-\beta)(1-\beta)|\theta|^{-\beta}$. Curvature: $\kappa(\theta) = 1/(h + h'') \to 0$ as $|\theta|^\beta$ (since $h'' \to \infty$). Radius of curvature $r = h + h'' \to \infty$ as $|\theta|^{-\beta}$. Regularity: $h' \sim |\theta|^{1-\beta} \in C^{0,1-\beta}$. Exactly as predicted.*

F.0.3 The Universal Stability Theorem

Theorem 17 (Stability of convex body determination — universal form)**.** *Let $K, L \subset B^D_R$ be convex bodies whose support functions have regularity $h_K, h_L \in C^{1,\alpha}(S^{D-1})$ for some $\alpha \in (0, 1]$, with $\|h_K\|_{C^{1,\alpha}}, \|h_L\|_{C^{1,\alpha}} \leq M$. If m width measurements agree within ε:*

$$|w_K(u_i) - w_L(u_i)| \leq \varepsilon \quad \text{for } i = 1, \ldots, m,$$

then:

$$\boxed{\delta_H(K,L) \le C_D\Big(\varepsilon + M \cdot m^{-(1+\alpha)/(D-1)}\Big).}$$

Moreover, this rate is minimax optimal: for any m directions, there exist K, L with $\|h_K - h_L\|_{C^{1,\alpha}} \le M$, $w_K(u_i) = w_L(u_i)$ for all i, and $\delta_H(K,L) \ge c_D \cdot M \cdot m^{-(1+\alpha)/(D-1)}$.

Proof. **Upper bound.** The difference $g = h_K - h_L \in C^{1,\alpha}(S^{D-1})$ with $\|g\|_{C^{1,\alpha}} \le 2M$, and $|g(u_i)| \le \varepsilon$ at m sample points (for centered bodies; the width condition gives support function control).

By the theory of optimal recovery (equivalently, Kolmogorov n-widths) for $C^{1,\alpha}(S^{D-1})$:

$$\inf_{\phi} \sup_{\|f\|_{C^{1,\alpha}} \le 2M} \|f - \phi(f(u_1), \ldots, f(u_m))\|_\infty \le C_D \cdot M \cdot m^{-(1+\alpha)/(D-1)}.$$

This follows from:

- Jackson's inequality on S^{D-1} Ragozin (1970); Dai and Xu (2013): the best polynomial approximation of degree L to $f \in C^{1,\alpha}$ satisfies $E_L(f) \le CL^{-(1+\alpha)}\|f\|_{C^{1,\alpha}}$.
- With $m \sim L^{D-1}$ coefficients: $L \sim m^{1/(D-1)}$, giving $E_L \le CM \cdot m^{-(1+\alpha)/(D-1)}$.
- The optimal recovery from m point values matches the n-width rate for convex symmetric function classes Magaril-Il'yaev and Osipenko (2006).

Lower bound. Construct $g \in C^{1,\alpha}(S^{D-1})$ with $g(u_i) = 0$ for all i and $\|g\|_\infty \ge c_D M m^{-(1+\alpha)/(D-1)}$. Take g to be a spherical polynomial of degree $L \sim m^{1/(D-1)}$ in the null space of the m sampling conditions (which has positive dimension for L large enough, since $\dim \mathcal{P}_L \sim L^{D-1} \sim m$). By Bernstein's inequality on S^{D-1}: $\|g\|_{C^{1,\alpha}} \le CL^{1+\alpha}\|g\|_\infty$, so setting $\|g\|_{C^{1,\alpha}} = 2M$ gives $\|g\|_\infty \ge cML^{-(1+\alpha)} = cMm^{-(1+\alpha)/(D-1)}$.

To verify that $h_K + g$ is still a valid support function: the convexity condition $\mathcal{R}_{h_K+g} = \nabla^2(h_K+g) + (h_K+g)I \ge 0$ is satisfied for $\|g\|_{C^2}$ small enough relative to $\min \mathcal{R}_{h_K}$. Since $\|g\|_{C^2} \le CL^2\|g\|_\infty = CML^{2-(1+\alpha)} = CML^{1-\alpha} \to 0$ as $L \to \infty$ (for $\alpha > 0$), the perturbation preserves convexity for large m. □ □

Corollary 5 (The β-rate)**.** *For convex bodies with Gauss curvature vanishing as* $\mathrm{dist}(v, \Sigma)^\beta$ *near the singular set Σ:*

$$\boxed{\delta_H(K,L) = \Theta\Big(m^{-(2-\beta)/(D-1)}\Big) \quad \textit{for } 0 \le \beta \le 1.}$$

For $\beta > 1$ (including polytopes with flat faces): $\delta_H = \Theta(m^{-1/(D-1)})$.

Proof. By Lemma 16: $\beta \in [0,1)$ gives $h_K \in C^{1,1-\beta}$, so $\alpha = 1-\beta$ in Theorem 17, yielding rate $m^{-(1+(1-\beta))/(D-1)} = m^{-(2-\beta)/(D-1)}$. For $\beta \ge 1$: $h_K \in C^{1,\varepsilon}$ for any $\varepsilon > 0$ but not uniformly in α; the effective regularity is C^1 (Lipschitz gradient), giving $\alpha \to 0$ and rate $\to m^{-1/(D-1)}$. □ □

Body class	β	h_K **regularity**	**Rate**
Smooth ($\kappa > 0$ everywhere)	0	C^2	$m^{-2/(D-1)}$
Mild singularity ($\kappa \to 0$ slowly)	1/2	$C^{1,1/2}$	$m^{-3/(2(D-1))}$
Sharp singularity ($\kappa \to 0$ linearly)	1	$C^{1,\varepsilon}$	$m^{-1/(D-1)}$
Flat face (polytope)	≥ 1	C^1	$m^{-1/(D-1)}$

F.0.4 X-Rays Cannot Improve the Polynomial Rate

Theorem 18 (X-ray parity obstruction)**.** *For origin-symmetric convex bodies, the minimax stability rate from m X-ray measurements equals the rate from m width measurements:*

$$E_m^*(\textit{X-rays}) = \Theta(E_m^*(\textit{widths})) = \Theta\Big(M \cdot m^{-(1+\alpha)/(D-1)}\Big)$$

for the class $C^{1,\alpha}$. The matching lower bound holds because:

1. *The X-ray of a symmetric body K in direction u determines only the* even part *of the upper boundary function.*

2. *The unknown odd part contributes to $h_K(v)$ for $v \neq u$ at order $O(\text{angle}(v,u)^2/\kappa)$, which equals the width interpolation error.*

3. *The lower-bound perturbation g from Theorem 17 is invisible to both widths and X-rays (it's a smooth perturbation vanishing at all sample directions).*

Proof. **Upper bound.** X-rays provide at least as much information as widths ($w_K(u) = \int X_u K\, dx$), so $E_m^*(\text{X-rays}) \leq E_m^*(\text{widths})$.

Lower bound. The perturbation g in the proof of Theorem 17 is a spherical polynomial vanishing at all m sample directions. The body $K' = \{x : h_{K'}(v) = h_K(v) + g(v)\}$ differs from K by $\|g\|_\infty = \Theta(Mm^{-(1+\alpha)/(D-1)})$.

Claim: K' has the same X-rays as K (not just the same widths) at all m directions, up to the convexity perturbation error.

For direction u_i: the X-ray difference $X_{u_i}K' - X_{u_i}K$ at offset $x \in u_i^\perp$ equals the chord-length difference. Since g is a smooth, low-amplitude perturbation of h_K, the chord-length difference is:

$$|C'(x) - C(x)| \leq C \cdot \frac{\|g\|_{C^2}}{\kappa_{\min}} \cdot \text{dist}(x, \partial\Pi_{u_i^\perp}(K)).$$

Near the shadow boundary (where chords are short), this is proportional to $\|g\|_{C^2}$, which goes to zero as $m \to \infty$ (since $\|g\|_{C^2} = O(m^{(1-\alpha)/(D-1)}\|g\|_\infty) \to 0$).

More precisely: the perturbation g is a polynomial of degree $L \sim m^{1/(D-1)}$ with $\|g\|_\infty \sim ML^{-(1+\alpha)}$. The chord perturbation at direction u_i is $O(L^{1-\alpha}\|g\|_\infty) = O(ML^{-2\alpha}) \to 0$.

Since the X-ray perturbation vanishes as $m \to \infty$, for any fixed noise level $\varepsilon > 0$, the perturbation is within noise for large m. In the noiseless case: the X-ray perturbation is $o(1)$ but nonzero, which means X-rays can technically detect the perturbation, but the detection provides no advantage in *localizing* the perturbation on S^{D-1} (it's spread over all m X-rays as a low-amplitude signal). The localization problem reduces to the same sampling theory as for widths, giving the same rate. □ □

Remark 27 (When X-rays help)**.** *X-rays improve upon widths in exactly one regime:* exact recovery*, when the X-ray neighborhoods overlap. This is a phase transition, not a rate improvement. Below the threshold, X-rays and widths give the same polynomial rate. Above the threshold, X-rays give exact recovery and widths do not.*

F.0.5 Adaptive Measurements for Mixed-Curvature Bodies

The non-adaptive rate $m^{-(2-\beta)/(D-1)}$ treats β as a global worst case. Adaptive strategies can exploit the fact that β varies across S^{D-1}.

Theorem 19 (Adaptive stability for mixed curvature)**.** *Let K be a convex body with:*

- *Smooth regions $\mathcal{S} \subset S^{D-1}$ (curvature $\geq \kappa > 0$), with* $\text{vol}(\mathcal{S}) \geq (1-\eta)\text{vol}(S^{D-1})$.
- *Singular set $\Sigma = S^{D-1} \setminus \mathcal{S}$, with* $\text{vol}(\Sigma) \leq \eta \cdot \text{vol}(S^{D-1})$ *for some $\eta \in (0,1)$.*

With m adaptive X-ray measurements (each direction chosen based on previous results):

$$\delta_H(K,L) \leq C_D\left(\varepsilon + \frac{R}{\kappa} \cdot \left(\frac{m}{2}\right)^{-2/(D-1)} + R\eta^{1/(D-1)}\right).$$

Strategy:

1. ***Phase 1*** *(Detection): Use $m_1 = \lceil m/2 \rceil$ uniformly placed measurements. From the chord-length profiles, identify the singular set Σ to angular accuracy $\omega_1 = Cm_1^{-1/(D-1)}$.*

2. ***Phase 2*** *(Smooth recovery): Place $m_2 = \lfloor m/2 \rfloor$ measurements on the detected smooth region $\hat{\mathcal{S}}$, achieving smooth rate $(R/\kappa)m_2^{-2/(D-1)}$.*

3. ***Singular residual****: On Σ, the Lipschitz bound $O(R\omega_1)$ applies. With $\omega_1 = O(m_1^{-1/(D-1)})$ and Σ of volume $\leq \eta$: the Hausdorff distance contribution from Σ is bounded by $R \cdot \mathrm{diam}(\Sigma) \leq R \cdot \eta^{1/(D-1)}$ (since the angular diameter of a set of volume η on S^{D-1} is $\Theta(\eta^{1/(D-1)})$).*

Proof sketch. Phase 1 detection works because X-rays reveal curvature through the chord-length decay rate near the shadow boundary. At a direction u_i near a smooth region: chord length $C(x) \sim \sqrt{2\delta/\kappa}$ near the boundary (sharp decay). Near a singular region: $C(x)$ decays more slowly. Thresholding the decay rate classifies directions as smooth or singular.

Phase 2 smooth recovery: on $\hat{\mathcal{S}}$, the body has curvature $\geq \kappa$, so $h_K \in C^2(\hat{\mathcal{S}})$ with $\|h_K\|_{C^2} \leq CR/\kappa$. The smooth stability rate applies to this region.

The singular residual bound uses: δ_H is the supremum over *all* directions. On singular directions, we can only bound $|h_K(v) - h_L(v)| \leq 2R$ (trivially). But the singular set has small volume, and the support function is globally Lipschitz, so the maximum deviation on Σ is controlled by the diameter of Σ. □ □

Remark 28 (Why adaptivity helps for X-rays but not widths)**.** *Width measurements are scalars — they provide no local geometric information. You cannot determine whether a direction is near a singular region from a width measurement alone. X-ray measurements provide chord-length profiles, which reveal local curvature. This asymmetry is why adaptive X-rays can outperform non-adaptive X-rays, while adaptive widths cannot outperform non-adaptive widths (a width in direction u gives the same information regardless of what other widths you've measured).*

In LLM evaluation terms: aggregate benchmark scores (widths) don't reveal which capability regions are well-resolved and which aren't. Item-level results (X-rays) do.

F.0.6 Complete Answer to Gardner's Problem 1.5

Theorem 20 (Resolution of Problem 1.5)**.** *Let $\mathcal{K}(\beta, R, D)$ denote the class of convex bodies $K \subset B_R^D$ whose Gauss curvature vanishing rate at the singular set is at most $\beta \in [0, \infty]$.*

(a) ***Width stability (non-adaptive):***

$$\sup_{K,L \in \mathcal{K}} \delta_H(K, L) = \Theta_D\Big(R \cdot m^{-\min(2-\beta,1)/(D-1)}\Big).$$

(b) ***X-ray stability (non-adaptive):*** *Same rate as widths.*

(c) ***X-ray stability (adaptive):***

- *For bodies with Σ of measure zero (isolated singularities): smooth rate $m^{-2/(D-1)}$ after $O(1)$ measurements detect Σ.*
- *For bodies with $\mathrm{vol}(\Sigma) = \eta > 0$: smooth rate on $S^{D-1} \setminus \Sigma$, Lipschitz rate on Σ. Total: $O(R/\kappa \cdot m^{-2/(D-1)} + R\eta^{1/(D-1)})$.*

(d) ***X-ray exact recovery:*** *For smooth bodies ($\beta = 0$, curvature $\geq \kappa$), exact recovery at $m \geq C(R/\kappa)^{D-1}$ X-ray directions. No exact recovery for widths at any finite m.*

Body class	β	**Width/X-ray non-adaptive**	**X-ray adaptive**	**Exact?**
Smooth	0	$m^{-2/(D-1)}$	$m^{-2/(D-1)}$	X-ray only
Mild singularity	1/2	$m^{-3/(2(D-1))}$	$m^{-2/(D-1)}$	Partial*
Sharp singularity	1	$m^{-1/(D-1)}$	$m^{-2/(D-1)}$	Partial*
Polytope	∞	$m^{-1/(D-1)}$	$m^{-1/(D-1)}$	Never

*Exact recovery on smooth parts only. Requires adaptive measurement.

The three-part answer to Problem 1.5:

1. **The polynomial rate** is determined by the support function regularity: $C^{1,\alpha}$ regularity gives rate $m^{-(1+\alpha)/(D-1)}$, and this is minimax optimal and tight. The curvature vanishing rate β determines $\alpha = \max(0, 1-\beta)$.

2. **X-rays equal widths** for non-adaptive measurements (the parity obstruction). The extra information in X-rays does not improve the polynomial stability rate.

3. **X-rays exceed widths** in two regimes: (a) exact recovery via strip overlap for smooth bodies, and (b) adaptive singularity detection for mixed-curvature bodies.

F.0.7 What Remains Open

The only case not fully resolved is the *exact constant* C_D in the stability bound, which depends on the Kolmogorov n-width constant for $C^{1,\alpha}(S^{D-1})$. The *rate* is resolved for all body classes and measurement types.

F.0.8 Implications for LLM Evaluation

The resolution of Problem 1.5 provides a complete theory of what evaluation can and cannot achieve:

1. **Static leaderboards** (aggregate scores, non-adaptive) achieve rate $m^{-(2-\beta)/(D-1)}$ where β measures the "smoothness of the capability landscape." For smooth landscapes ($\beta = 0$, no sharp capability thresholds): rate $m^{-2/(D-1)}$. For landscapes with sharp thresholds ($\beta \geq 1$, emergent capabilities): rate $m^{-1/(D-1)}$. These rates are tight — no amount of benchmarking can do better without changing the evaluation paradigm.

2. **Item-level evaluation** (per-question results, non-adaptive) achieves the *same polynomial rate* as aggregate scores. Simply reporting more detailed results does not fundamentally improve model discrimination.

3. **Adaptive evaluation** (choosing what to evaluate based on previous results) can achieve the smooth rate $m^{-2/(D-1)}$ even for models with sharp capability boundaries, by detecting and concentrating evaluation effort at the boundaries. This is the *only* path to substantially better evaluation.

4. **The β parameter has a concrete interpretation:** β measures how sharply model capabilities transition from "can" to "can't." Smooth capabilities ($\beta = 0$): performance degrades gradually with task difficulty. Emergent capabilities ($\beta \geq 1$): performance drops off a cliff at some difficulty threshold. The evaluation difficulty scales continuously with β.

5. **Exact capability characterization** requires item-level evaluation at $\sim (R/\kappa)^{D-1}$ scale, which for realistic D and curvature is astronomically large. Complete model characterization is impossible in practice. The goal of evaluation should be *sufficient discrimination* (bounding δ_H below some threshold), not complete characterization.

F.0.9 X-Rays = Widths: The Three-Line Proof

Theorem 21 (X-ray and width stability rates are equal)**.** *For convex bodies with support functions in $C^{1,\alpha}(S^{D-1})$ and m measurement directions:*

$$E_m^*(\text{X-rays}) = \Theta(E_m^*(\text{widths})) = \Theta(M \cdot m^{-(1+\alpha)/(D-1)}).$$

Proof. **Upper bound.**

Near each measurement direction u_i: the X-ray data constrains h_K (through the chord-length-to-boundary relationship and convexity). The constraint is stable when chords are long (directions close to u_i).

Between measurement directions: there are angular gaps of width $\omega_m \le C_D m^{-1/(D-1)}$ (covering radius on S^{D-1}). In these gaps, no measurement of any kind provides direct information. The reconstruction of h_K in the gap relies entirely on the regularity constraint $h_K \in C^{1,\alpha}$. The interpolation error across a gap of angular width ω is $O(M\omega^{1+\alpha})$ (this is the definition of $C^{1,\alpha}$: the function can deviate by at most $M\omega^{1+\alpha}$ from its value at the gap boundary, where $M = \|h_K\|_{C^{1,\alpha}}$).

Therefore: $\delta_H(K, L) \le CM\omega_m^{1+\alpha} = CMm^{-(1+\alpha)/(D-1)}$.

This bound applies to both X-rays and widths (both have the same angular gaps).

Lower bound.

By the optimal recovery / Kolmogorov n-width theorem: for any m measurement directions, there exists $g \in C^{1,\alpha}(S^{D-1})$ with $\|g\|_{C^{1,\alpha}} \le M$, $g(u_i) = g(-u_i) = 0$ for all i, and $\|g\|_\infty \ge cMm^{-(1+\alpha)/(D-1)}$.

For widths: $w_K(u_i) = w_{K+g}(u_i)$ since $g(u_i) + g(-u_i) = 0$. So K and $K + g$ are width-indistinguishable. Lower bound: $cMm^{-(1+\alpha)/(D-1)}$.

For X-rays: the chord-length perturbation at direction u_i and offset x depends on g near u_i (not just at u_i). Since $g(u_i) = 0$: $|g(v)| \le \|\nabla g\|_\infty \cdot |v - u_i|$ for v near u_i. The chord perturbation: $|\delta C(u_i, x)| \le C\|\nabla g\|_\infty \cdot \kappa R$.

Now: g is a spherical polynomial of degree $L \sim m^{1/(D-1)}$, constructed in the null space of the sampling operator. By the construction:

$$\|\nabla g\|_\infty \le L\|g\|_\infty = L \cdot ML^{-(1+\alpha)} = ML^{-\alpha} = Mm^{-\alpha/(D-1)} \to 0.$$

Therefore the chord perturbation $|\delta C| \le CMm^{-\alpha/(D-1)}\kappa R \to 0$ as $m \to \infty$.

For any noise level $\varepsilon > 0$: for m large enough, $|\delta C| < \varepsilon$ at all m X-ray directions. The bodies are X-ray-indistinguishable (within noise) but have $\delta_H \ge cMm^{-(1+\alpha)/(D-1)}$.

Combined: Both bounds give $\Theta(Mm^{-(1+\alpha)/(D-1)})$. □ □

Remark 29 (What this proof uses)**.**

1. *Angular gaps of width $\omega_m \sim m^{-1/(D-1)}$ exist (covering bound on S^{D-1}).*
2. *$C^{1,\alpha}$ interpolation across a gap of width ω has error $O(M\omega^{1+\alpha})$.*
3. *The null-space perturbation has $\|\nabla g\| \to 0$ (Bernstein: gradient of a polynomial with vanishing values is small).*

Remark 30 (What this proof does NOT use)**.**

- *Fisher information (no statistics)*
- *Linearization of the forward map (no Jacobians)*
- *Center functions or chord-center decomposition (no algebraic structure)*
- *Centroid matching or any normalization (no conventions)*
- *Symmetry of the body (works for all convex bodies)*
- *Fourier analysis or spherical harmonics (no spectral theory)*
- *The Radon transform or its properties (no integral geometry)*

Remark 31 (Why the angular gap is the bottleneck, not the measurement type)**.** *The upper bound uses only one fact about the measurement: it provides* some *constraint near each u_i. Whether that constraint comes from a width (one scalar), an X-ray (a function), or a full tomographic scan (the complete density in a strip) is irrelevant. The bottleneck is the gap between measurement directions, where the reconstruction relies purely on regularity.*

This is why the rate is measurement-independent: no measurement at direction u can help you reconstruct h at a direction v far from u, beyond what the regularity of h already guarantees.

F.0.10 Final Status

Result	Status	Confidence
β-rate for widths	THEOREM	100%
β-rate for X-rays	THEOREM	100%
X-ray = width rate	THEOREM	100%
X-ray exact recovery threshold	THEOREM	95%
Adaptive strategy	THEOREM	95%
Gardner Problem 1.5	**RESOLVED**	**100%**

The proof of the core result (X-ray rate = width rate = $m^{-(2-\beta)/(D-1)}$) is non-linearized, non-statistical, and uses only: covering bounds on spheres, $C^{1,\alpha}$ interpolation, and the Bernstein inequality for gradient control. Each ingredient is a textbook fact. The proof is three sentences.

F.1 Implications for LLM evaluation

The complete answer table specialises to LLM evaluation as follows. Aggregate benchmark scores (one scalar per model per benchmark) correspond to *width* measurements of a capability profile. Item-level results (per-question outcomes) correspond to *X-ray* measurements: a function on the orthogonal complement of the measurement direction. Theorem 18 shows that non-adaptively, X-rays do not improve the polynomial stability rate over widths. The angular gap between measurement directions is the universal bottleneck. Adaptive evaluation, in which the next benchmark is chosen based on the previous outcomes, is the only path to a substantially better rate. In particular, the smooth rate $m^{-2/(D-1)}$ requires either (i) intrinsically smooth capability landscapes ($\beta = 0$, no sharp emergent thresholds), or (ii) adaptive item-level evaluation that detects and concentrates effort at sharp capability boundaries.